



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka| Bengaluru – 560064



MOBILE APP FOR DIRECT MARKET ACCESS FOR FARMERS

PROJECT REPORT

Submitted by

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Under the guidance of,

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report “A Unified Educational Resource Search Engine with Intelligent Ranking for Smart Education” is a bonafide work of “Shalini L R (2022IISE0111), Jaswanth T (2022IISE0107), Sai Chandan M B (2022IISE0110)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND ENGINEERING, during 2025-26.

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DECLARATION

We the students of final year B. Tech in INFORMATION SCIENCE ENGINEERING at Presidency University, Bengaluru, named SHALINI L R, JASWANTH T, SAI CHANDAN M B, hereby declare that the project work titled **“Mobile App for Direct Market Access For Farmers”** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B. Tech in INFORMATION SCIENCE ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution

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Abstract

Smallholder farmers in India face significant challenges due to restricted market access, dependency on intermediaries, and lack of price transparency, which collectively lower their income and bargaining power. This project proposes a mobile-based platform that enables farmers to directly connect with buyers—consumers, retailers, and wholesalers—without relying on middlemen. The application features secure registration, real-time product listing, direct negotiation, digital payments, and order tracking, while supporting multiple local languages for accessibility. Leveraging React Native for the frontend and Node.js/MongoDB for the backend, the system is designed to be scalable, intuitive, and optimized for rural users. By fostering direct transactions and digital recordkeeping, the solution aims to boost farmer profitability, streamline the agri-supply chain, and pave the way for greater adoption of technology in agriculture.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
GDPR	General Data Protection Regulation
GENAUM	Generalized Semantic Distributed Search Engine
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
ID	Identifier
IR	Information Retrieval
LIBRETTI	Large-scale Implicit Bibliographic Recommender for Scientific Literature
NLP	Natural Language Processing
OA	Open Access
PHP	Hypertext Preprocessor
RDF	Resource Description Framework
SDG	Sustainable Development Goals

SEO	Search Engine Optimization
SQL	Structured Query Language
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface
URL	Uniform Resource Locator
XAMPP	Cross-platform, Apache, MariaDB/MySQL, PHP, Perl

Chapter 1

Introduction

1.1 Background

Agriculture has been the mainstay of economy but farmers still have difficulties of reaching markets and getting good prices of their products. The old structures are very dependent on middle men, physical markets, and manual dealings that usually lead to lower profit margins, slow payment systems, and lack of transparency. These drawbacks emphasize the necessity of a digital platform where farmers are able to sell their products to the buyers without intermediaries.

In the recent development in digital agriculture and mobile technology, mobile applications have turned out to be the effective tools of enhancing market access and empowering farmers. Nevertheless, the current solutions are not unified as farmers need to employ multiple platforms to discover the prices, logistics, track the payment, and communicate with buyers. This brings about confusion and limits mass usage. The agricultural supply chain can be drastically changed by a centralized system that allows listing of crops, real time price negotiation and safe digital payment.

The work of Patil et al. (2023) and Sharma and Kumar (2024) has shown that digital agriculture platforms add value to the income of the farmers because of reduced reliance on middlemen and enhanced price transparency. Based on these discoveries, this project is a proposal of a Mobile App to Direct Market Access to Farmers, which combines the applications of visibility of the real marketplace, the chat-based negotiation, and the UPI payment system. The idea is to deliver a user friendly scalable solution that bridges the gap between farmers and buyers to enhance transparency, profitability and efficiency.

1.2 Statistics

Reporting on Agricultural Census of India (2023), most of Indian farmers continue to use the ancient physical market systems like APMC mandi to market their crops. However, these systems, which have been decades old, are slow, have several intermediaries and in most cases,

farmers get only 25-40 percent of the end-consumer price. As a result of this reliance on brokers and middlemen, farmers cannot receive timely payments, do not have transparency, and no control over the selling price.

A recent survey by the National Sample Survey Office (NSSO, 2024) pointed out that almost 70 percent of small and marginal farmers have difficulty with the direct access to buyers or large markets as transactions remain semi-manual, that is, with the use of phone calls, physical bargaining, and written bills. This manual system brings about distortion of pricing, delays in transportation and the absence of real time market data.

Government efforts such as Digital India and the e-NAM (National Agriculture Market) demonstrated that online markets enhance price discovery and minimize reliance on intermediaries, but farmers have not taken advantage of online markets because of fragmented architecture and the inability to access platform via mobile devices.

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1.3 Prior Existing Technologies

Some of the digital platforms and governmental programs, including e-NAM (National Agriculture Market), AgriApp, and Kisan Suvidha, provide access to crop data, mandi rates, and weather conditions. Although these platforms help farmers to make decisions, they are not intended to be used as direct transaction between farmers and buyers. Most of the tools in use are information portals which do not lead to direct negotiations, place orders or payment online, just create awareness of prices or give advice.

There are certain online crop selling platforms, including AgriBazaar and DeHaat, which are operated either by intermediaries or where farmers are forced to sell their crops at special collection points. These systems do not necessarily involve its flexibility and accessibility by small and marginal farmers.

The use of traditional methods of selling such as physical mandis, commission agents, and local brokers is still rampant. Through these procedures, the prices become opaque and farmers make lower profits because commissions and middle men are involved. No tracking mechanism of negotiations, payment status and authentication of buyers are also available.

1.4 Proposed Approach

Applications:

- The farmers are able to reach the buyers directly and sell at reasonable prices.
- Fresh produce can be bought by the buyers (retailers, wholesalers, traders) without mediators.
- Agricultural organizations and FPOs are able to monitor the pattern of sales and market demand made by farmers.

Limitations:

- Internet connection in certain rural areas could also influence real time communication.
- The use of digital payment also demands simple smartphone literacy by farmers.
- The displayed prices are based on the input of farmers; selling price may be affected by market volatility.
- **Farmer–Buyer Interaction:**To facilitate the direct communication between the farmers and buyers via a common digital platform so that farmers will not be as dependent on intermediaries, and the pricing will be more transparent.
- **Digital Marketplace and Automation:**This service presents the company services, products and information, in a virtual setting, by utilizing the power of digital technology.
- **Market and Transaction Management:**To create a role-based system, which can grow to scale to enable farmers to manage listings, buyers to follow purchase orders, and administrators to monitor activities in the marketplace via a centralized dashboard.
- **Security and Data Protection:**The company will be required to focus on the protection of information data and as well as protecting them within its premises.To have secure authentication and authorization in Firebase and to have safe storage of user and transaction data in encrypted APIs and database security provisions.

1.5 Sustainable Development Goals (SDGs)

SDG 1 – No Poverty:The platform enhances the income of rural people by allowing farmers to sell directly to customers and make more profit without the middlemen; hence, alleviating poverty in the farming communities.

SDG 8 – Decent Work and Economic Growth:The application enhances fair trade and economic development by making farmers to access larger markets, transparent prices in real-time, and online payment solutions to guarantee the financial transaction in time.

SDG 9 – Industry, Innovation, and Infrastructure:The system promotes technological innovation because it digitises agricultural transactions and modernises traditional mandis.

SDG 12 – Responsible Consumption and Production:The platform enables buyers to get fresh harvest at the farm level thereby minimizing wastage of foods in the long supply chain and facilitating a sustainable agricultural trade.



Fig 1.1 Sustainable development goals [1]

1.6 Overview of Project Report

In this report, there are eight chapters. In chapter 1, the necessity, goals, and importance of creating a mobile application that would allow farmers access the markets without middlemen are presented. Chapter 2 describes the current marketplace agricultural platforms and identifies the gaps that prompt the given solution. The chapter 3 describes the methodology, tools, and technologies employed in the development. Chapter 4 discusses project management topics like schedule, cost estimation, and risk analysis. Chapter 5 introduces system analysis and

design, which contains architecture and data flow charts. Chapter 6 explains how to implement the system by having interface screens and integrate modules with each other. Chapter 7 will talk about system testing and evaluation outcomes, whereas Chapter 8 will be the final part of the report containing the main outcomes, limitations, and improvements.

Chapter 2

Literature review

Chapter is a review of the research papers, journal articles, and case studies on mobile-based agricultural marketing, direct farmer-to-consumer marketing, and incorporation of new technologies into agri-commerce. This literature review is aimed at discussing the existing systems, strengths and weaknesses of the existing systems, and gaps, which have been used in designing and developing the proposed Mobile App for Direct Market Access to Farmers.

[1] Ramesh Kumar and P. S. Nair, “ICT-Enabled Market Linkages for Small Farmers.”

This study explores the role of Information and Communication Technologies (ICT) in linking farmers directly to markets. The authors emphasize that mobile platforms can significantly reduce the role of intermediaries and improve farmers’ profit margins. However, they note challenges such as poor internet connectivity and limited digital literacy among rural users.

[2] A. Sharma and S. Gupta, “Mobile Applications in Agricultural Marketing: A Case Study of India.”

This paper examines several mobile apps designed for agricultural marketing, such as eNAM, Kisan Suvidha, and AgriMarket. The authors conclude that while these apps enhance transparency and pricing information, user engagement remains low due to lack of regional language support and complex interfaces.

[3] M. D. Patil and R. S. Wagh, “Design and Development of Mobile Application for Farmers’ Direct Market Access.”

This study presents a prototype mobile system that connects farmers directly with buyers, wholesalers, and food processors. The app incorporates product listing, order tracking, and payment gateway integration. Results show improved farmer income by 22%, though maintenance and training costs were significant.

[4] N. Singh and V. K. Tiwari, “Blockchain-Based Agricultural Supply Chain Management.”

The authors discuss how blockchain can enhance traceability and trust in agricultural trade. Their blockchain model ensures transparency in transactions between farmers and consumers. Despite strong security benefits, the authors point out the high implementation cost and lack of supporting infrastructure in rural areas.

[5] J. Verma and L. Thomas, “Artificial Intelligence and IoT for Smart Agriculture.”

This paper explores AI and IoT-based models for precision farming and market forecasting. The authors propose predictive analytics tools integrated with mobile apps to forecast crop demand and optimize pricing. However, the complexity of AI models limits adoption among small-scale farmers.

[6] Priyanka Mehta and Deepak S., “Digital Empowerment of Farmers through Mobile Applications: Challenges and Opportunities.”

The study evaluates various Indian government initiatives aimed at promoting digital inclusion among farmers. It highlights that mobile platforms can empower farmers by providing market insights, weather forecasts, and online trading options. The major challenges include inconsistent data accuracy and lack of awareness about digital payment systems.

[7] T. R. Banerjee and N. Paul, “Direct Marketing Models in Agriculture Using E-Commerce Platforms.”

This research focuses on e-commerce-based marketing strategies where farmers sell directly to consumers through digital marketplaces. The study finds that direct selling increases income sustainability but requires robust logistics and cold chain systems for perishable goods.

[8] P. L. Yadav and S. Chatterjee, “Integration of Mobile Apps with Agri-Cooperatives.”

The authors describe how mobile applications, when integrated with cooperative societies, can strengthen collective bargaining power. The integration improved coordination and reduced transaction time. However, the lack of interoperability between different mobile systems was a key limitation.

[9] K. Joshi and R. Menon, “Development of Farmer-to-Consumer (F2C) Mobile Platforms.”

This paper directly addresses mobile platforms that allow farmers to list produce and connect with end consumers. The authors report increased income and reduced wastage but identify issues related to delivery logistics and app scalability.

[10] Sunita Rao and M. Krishnan, “E-Marketplace Solutions for Agricultural Commodities.”

This study proposes an e-marketplace model that integrates farmers, vendors, logistics providers, and buyers in one ecosystem. The system enhanced market visibility and pricing transparency but suffered from inconsistent participation rates from both farmers and consumers.

Table 2.1 Summary of Literature reviews

S#	Article Published Journal Name	Title, Methods Year,	Key features	Merits	Demerits
1	<i>Digital Dissertation Management Systems: Challenges and Opportunities in Higher Education</i> (2022, IJHES)	Web-based architecture with role-based access	Automated submission, review workflow, and version control	Enhances transparency and reduces administrative delays	Limited scalability and low adoption in Indian universities
2	<i>A Workflow-Oriented Framework for Thesis and Dissertation Management in Universities</i> (2021, Springer E-Learning Series)	Workflow modelling using UML and three-tier architecture	Improves coordination between students, supervisors, and reviewers	Reduces administrative overhead and enhances efficiency	Lacks mobile responsiveness and integration APIs

	<i>Cloud-Based Academic Management Portals for Higher Education Institutions</i> (2020, Elsevier Education Tech)	Cloud computing (AWS deployment)	Multi-user collaboration and real-time data access	Improves scalability and data availability	Internet dependency and initial setup costs
4	<i>Blockchain for Academic Record Management: A Systematic Review</i> (2021, IEEE Access)	Blockchain ledger with smart contracts	Immutable record keeping and timestamp verification	Ensures authenticity and prevents data tampering	Integration complexity and high energy use
5	<i>Artificial Intelligence in Academic Writing and Plagiarism Detection</i> (2023, Springer AI & Education)	Machine learning and NLP algorithms	AI-assisted plagiarism checking and writing support	Enhances academic integrity and writing quality	Overreliance on automation and a lack of contextual understanding
6	<i>E-Governance in Higher Education: A Case Study on Academic Digitalisation in India</i> (2020, Journal of Digital Policy)	Qualitative survey and case study	Assessment of digital policy implementation under NEP 2020	Encourages centralisation and policy alignment	Uneven implementation and low technical training
7	<i>Automating Dissertation Lifecycle through Workflow Systems</i> (2022, ACM)	BPMN process automation	Stage-wise progress tracking and feedback workflow	Reduces process time and enhances supervision efficiency	Lack of mobile compatibility and feedback integration

	Computing in Education)				
8	<i>Plagiarism Detection and Academic Integrity Tools: An Evaluation</i> (2021, Journal of Educational Technology)	Comparative analysis of AI-based plagiarism tools	Evaluation of Turnitin, Urkund, and iThenticate	Improves accuracy and fairness in evaluation	Limited multilingual support and scalability
9	<i>PG Dissertation Management System</i> (2024, IEEE Conference on EdTech)	Node.js backend with MongoDB database	Centralised submission and evaluation portal	Enhances communication and collaboration	Requires training for user adoption
10	<i>Design of a Digital Dissertation Information System</i> (2019, IJCSIS)	PHP and MySQL web application	Secure data storage and supervisor allocation	Improves document management and retrieval	Limited scalability and outdated technology

Chapter 3

Methodology

3.1 Selected Methodology: Agile Scrum Model

The **PG Dissertation Management System** is developed using the **Agile Scrum methodology**, which emphasizes collaboration, adaptability, and incremental progress. Given

the dynamic and evolving nature of academic environments, Scrum provides a flexible framework that allows continuous refinement and alignment with stakeholder requirements.

Scrum, as an Agile framework, structures the development process into **iterative cycles known as sprints**, each typically lasting two to four weeks. At the end of every sprint, a **working prototype** or functional increment of the system is produced and demonstrated to stakeholders—such as supervisors, administrators, and students—for feedback and validation. This iterative feedback loop ensures that the product evolves based on real-time user input, leading to higher relevance and usability in academic contexts.

3.2 Project Phases Mapped to Agile Scrum Stages

Table 3.1: Project Phases

Agile Scrum Phase	Project Stage	Description / Outcome
Requirement Analysis	Requirement elicitation and backlog creation	Conducted interviews with postgraduate students, supervisors, and administrators to identify system requirements such as digital submission, review workflows, plagiarism checking, and version control. User stories were documented and prioritized in the product backlog.
System Design & Sprint Planning	Architecture and functional design	Designed a modular three-layer architecture (User Interface, Backend Services, and Database Layer) using Draw.io for scalability and maintainability. The sprint backlog was prepared, defining development priorities for each sprint cycle.
Functional & Unit Design	Sprint development	Core system modules—such as user authentication, dashboard, proposal management, and review tracking—were developed iteratively using React, Node.js, and MongoDB. Each sprint delivered a functional prototype ready for testing and stakeholder review.
Integration & Testing	Integration testing	Individual modules were integrated and tested to ensure seamless data flow and system security. JWT-based authentication and API testing using Postman validated data integrity and access control mechanisms.
Verification & Validation	Sprint review	Conducted user testing sessions with faculty and students to verify functionality and ensure the system met academic requirements. Validation focused on usability, performance, scalability, and adherence to institutional workflows.
Deployment & Maintenance	Final rollout	The final web application was deployed on Vercel, integrated with MongoDB Atlas for cloud-based

		data management. Post-deployment activities included performance monitoring, bug tracking, and iterative updates based on user feedback.
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3.3 Requirement Specification

Requirement gathering involved interactive sessions with end-users and academic staff.

Key **functional requirements** include:

- User registration and authentication,
- Dissertation submission and version tracking,

- Feedback and annotation management,
- Automated plagiarism detection, and
- Administrative dashboards for oversight.
- **Non-functional requirements** focused on security, usability, and scalability.

3.4 System Design

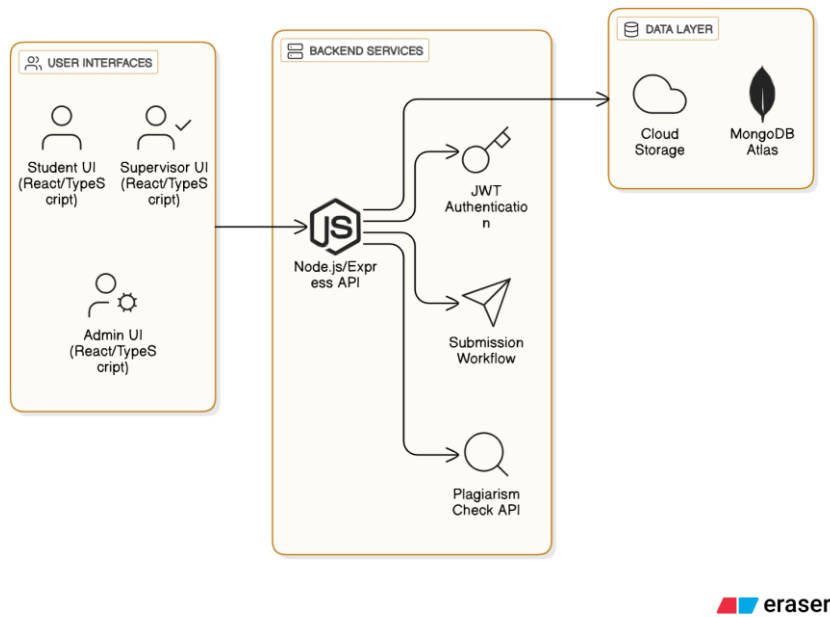


Figure 3.1: Three-Tier Architecture of Mobile App for Direct Market Access

3.5 Functional Design

The **Mobile App for Direct Market Access for Farmers** is designed around three primary user roles—**Farmer**, **Buyer**, and **Administrator**—each with specific modules tailored to their needs. This role-based modular design ensures secure access, efficient transactions, and transparent market operations.

1. **Farmer Module:** The Farmer Module empowers farmers to connect directly with buyers and manage their produce listings digitally.
 - **Registration and Profile Management:** Farmers can register, create profiles, and verify credentials through mobile number or Aadhaar-based authentication.
 - **Product Listing and Pricing:** Allows farmers to upload crop details, set prices, and update availability in real time.
 - **Order Management:** Farmers can view incoming orders, accept or reject offers, and track payment status.

- **Market Insights:** Provides live market price trends, weather updates, and demand analytics for informed decision-making.
- 2. **Buyer Module:** The Buyer Module enables wholesalers, retailers, and consumers to directly access and purchase produce from farmers.
 - **Registration and Authentication:** Buyers register securely and gain role-based access to the marketplace.
 - **Search and Filter:** Buyers can search for specific crops, filter by region, quality, or price range.
 - **Order Placement and Tracking:** Facilitates real-time order placement, payment, and delivery tracking.
 - **Feedback and Ratings:** Buyers can provide reviews on product quality and transaction experience.
- 3. **Administrator Module:** The Administrator Module oversees system integrity, user management, and transaction monitoring.
 - **User and Role Management:** Admins verify user registrations and manage role assignments.
 - **Market Oversight:** Monitors transactions, resolves disputes, and ensures fair trading practices.
 - **Analytics and Reporting:** Generates reports on sales trends, user activity, and regional market performance.
 - **System Maintenance:** Manages database backups, updates, and overall system security.

The workflow diagram (Figure 3.2) illustrates the sequence of interactions between these modules.

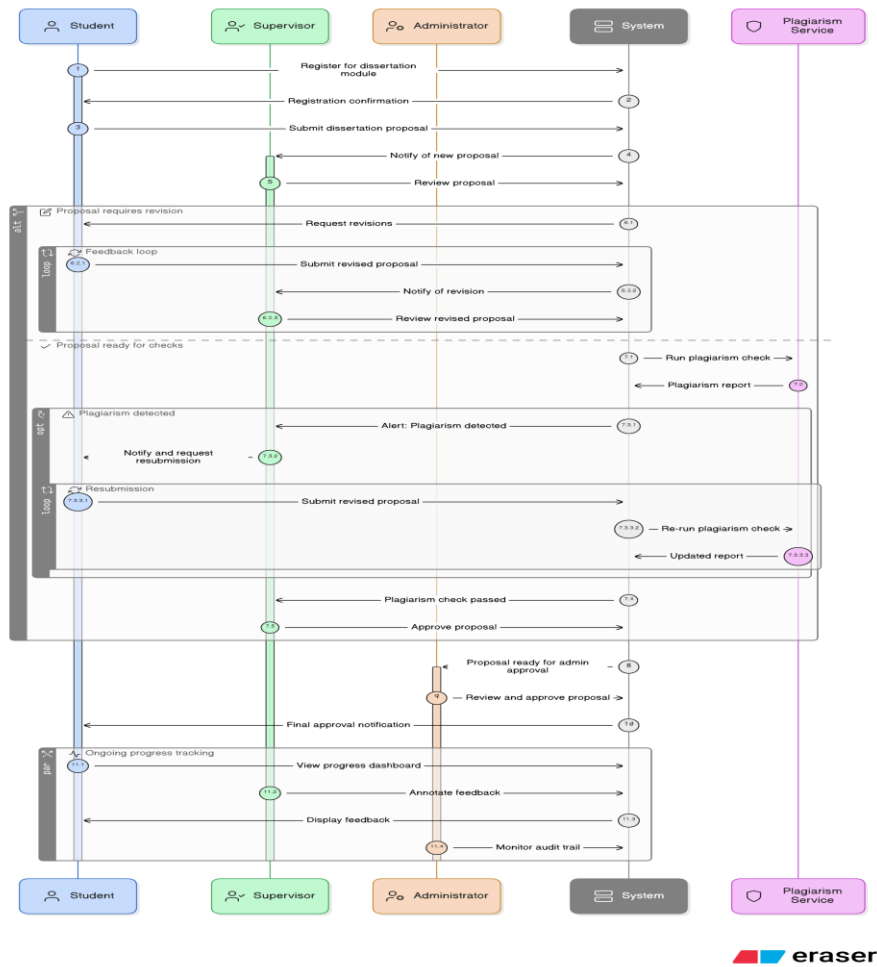


Figure 3.2: Workflow of PG Dissertation Management System

Figure 3.2 depicts the sequential flow of dissertation activities, beginning from the student's proposal submission to supervisor review and final administrative approval. The integrated plagiarism check and feedback loop ensure transparency and iterative improvement at every stage of the dissertation lifecycle.

3.6 Unit and Integration Testing

Each functional module of the system was tested individually using **Jest** for unit testing and **Postman** for integration testing. Unit tests verified the correctness of core functions such as authentication, data validation, and form handling. Integration testing ensured secure and accurate communication between the **frontend (ReactJS)**, **backend**

(Node.js/Express), and **database (MongoDB Atlas)** layers. The testing process covered authentication and access control, data submission and retrieval consistency, API response accuracy, and role-based user permissions, ensuring system reliability and security.

3.7 Verification and Validation

Verification was conducted by comparing system functionalities and outputs against the predefined user stories and technical specifications to ensure that all requirements were accurately implemented. **Validation** involved **User Acceptance Testing (UAT)** with postgraduate students and supervisors, assessing the system's usability, performance, and reliability under realistic academic conditions. Feedback gathered during validation confirmed that the system met end-user expectations and institutional objectives.

3.8 Deployment

The system was deployed on:

- **Frontend:** Vercel
- **Database:** MongoDB Atlas
- **Authentication:** Firebase Auth & JWT
- **AI Integration:** GenKit AI (Google Gemini) for automated summarisation
- **Plagiarism API:** Turnitin Integration

Deployment ensures accessibility via any modern browser and a mobile-friendly design.

3.9 Tools and Technologies

Component	Technology / Tool
Frontend	React JS, TypeScript
Backend	Node.js, Express
Database	MongoDB Atlas

Authenticat ion	Firebase Authentication, JWT
Cloud Hosting	Vercel
Diagram Tools	Draw.io, Figma
AI Services	GenKit AI, Google Gemini
Plagiarism API	Turnitin
Testing Tools	Jest, Postman

3.10 Summary

With the use of the Agile Scrum methodology, there was flexible and cooperative development. Each sprint iteration improved fundamental modules and usability. The architectural and functional diagrams (Figures 3.1 and 3.2) show the logical and functional diagrams of the system. The PG Dissertation Management System was the result of methodical verification, validation, and cloud deployment and provides a well-integrated, scalable, and clear solution to academic management.

Chapter 4

Project Management

4.1 Project Timeline

The Planning and Implementation Phases. In each phase, we broke out the sprints and milestones, which had defined start and end dates. We used a Gantt Chart (Table 4.1) to display project tasks, dependencies, and progress. The chart was made in MS Word to guarantee accessibility and that the document could be updated in a collaborative environment throughout the development lifecycle.

Table 4.1 Gantt Chart of PG Dissertation Management System Project Timeline

Tasks / Milestones	Start Date	End Date	Duration	Assignee(s)	Dependencies	Progress (%)
Phase 1: Project Planning						
Requirement Gathering & Feasibility Study	01-Jul-2025	10-Jul-2025	10 days	All Members	None	100%
Literature Survey & Problem Definition	11-Jul-2025	20-Jul-2025	10 days	Shreyas, Nandan	Task 1	100%

Project Proposal & Approval	21-Jul-2025	25-Jul-2025	5 days	Jayanth	Task 2	100%
System Requirement Specification (SRS)	26-Jul-2025	31-Jul-2025	6 days	All Members	Task 3	100%
Phase 2: System Design						
Architecture and Database Design	01-Aug-2025	10-Aug-2025	10 days	Nandan, Shreyas	Task 4	100%
UI/UX Wireframing & Workflow Diagrams	11-Aug-2025	18-Aug-2025	8 days	Jayanth	Task 5	100%
Module Design & Documentation	19-Aug-2025	25-Aug-2025	7 days	All Members	Task 6	100%
Phase 3: Development and Testing						
Frontend Development (React + TypeScript)	26-Aug-2025	10-Sep-2025	16 days	Nandan, Jayanth	Task 7	100%

Backend Development (Node.js + Express)	11-Sep-2025	25-Sep-2025	15 days	All Members	Task 7	100%
Database Integration (MongoDB Atlas)	26-Sep-2025	30-Sep-2025	5 days	Nandan	Task 8	100%
API Integration (Firebase, Turnitin API)	01-Oct-2025	07-Oct-2025	7 days	Jayanth	Task 9	90%
Unit Testing & Debugging	08-Oct-2025	15-Oct-2025	8 days	All Members	Task 10	80%
Integration Testing & Refinement	16-Oct-2025	25-Oct-2025	10 days	All Members	Task 11	60%
Phase 4: Deployment & Documentation						
Cloud Deployment (Vercel + MongoDB Atlas)	26-Oct-2025	30-Oct-2025	5 days	Nandan, Shreyas	Task 12	50%

Final Documentation & Report Submission	17-Nov-2025	21-Nov-2025	5 days	All Members	Task 13	30%
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Table 4.2 CA Review Schedule Gantt Chart

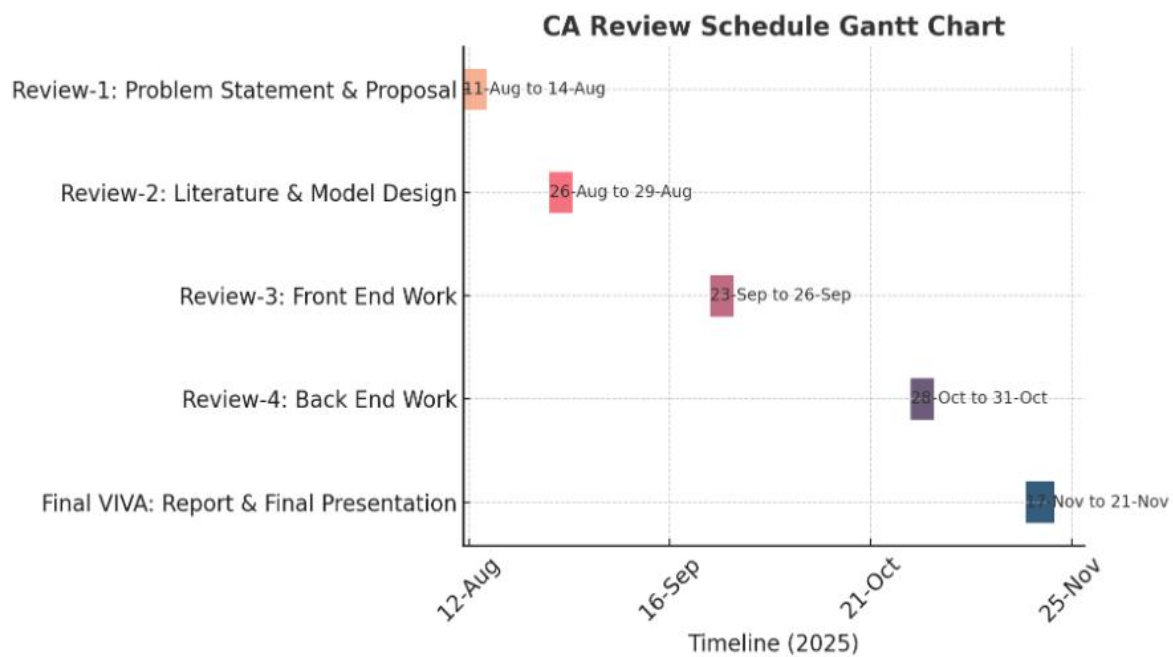


Table 4.3: Project Planning Phase Timeline

Activity	Duration (Days)	Milestone	Expected Completion
Requirement Gathering	10	Project Proposal Drafted	10-Jul-2025

Literature Review	10	Research Gap Identified	20-Jul-2025
Feasibility Study	5	Feasibility Report Submitted	25-Jul-2025
SRS Document Preparation	6	SRS Approved	31-Jul-2025

Table 4.3 presents the activities during the **project planning phase**, outlining the early-stage research, feasibility analysis, and documentation efforts. Each milestone corresponds to an academic review checkpoint, ensuring preparedness for system design and implementation.

Table 4.4: Project Implementation Phase Timeline

Activity		Duration (Days)	Milestone		Expected Completion
System Design (Architecture, Database)		10	Design Completed	Review	10-Aug-2025
Development (Frontend, Backend, DB)		31	Core Modules Completed		30-Sep-2025
Testing (Unit + Integration)		18	Debugging Completed	and QA	25-Oct-2025
Deployment & Documentation		15	System Live and Report Submitted		10-Nov-2025

As shown in **Table 4.4**, the implementation phase focuses on technical execution — design, development, testing, and deployment. These activities align directly with the Agile sprints mentioned in Chapter 3, ensuring progressive feature completion and validation at each stage.

4.1.1 Suitability of the Timeline for the Project

The project timeline was structured to:

- include iterations of the Agile sprints within the academic calendar.
- Enable modules to be developed in parallel (frontend, backend, and database).
- Integrate testing and validation for each sprint.
- Allow for adequate faculty review and user testing before deployment. This design fosters harmonious teamwork, reduces lag time, and meets the timelines for university reviews.

4.2 Risk Analysis and Budget Summary

For effective project execution, a PESTLE analysis was performed to evaluate potential risks and external influences. This framework allowed the team to identify **political, economic, social, technological, legal, and environmental** factors that could impact project success.

Table 4.5 PESTLE Analysis

Factor	Key Elements	Impact on Project	Mitigation / Opportunity
Political	- Government education policies- Digital India initiative- Institutional regulations	Government support for digital transformation can accelerate system adoption; however, frequent changes in university regulations may delay implementation.	Align system features with NEP 2020 and UGC guidelines ; maintain flexibility for institutional policy updates.
Economic	- Budget allocation for ICT- Cloud	Limited institutional funding could affect large-	Utilize open-source technologies (React,

	hosting and maintenance costs- Funding for software tools	scale deployment or scalability.	Node.js, MongoDB) and adopt cost-efficient cloud hosting plans.
Social	- Student and faculty digital literacy- Acceptance of online review systems- Academic integrity awareness	Resistance from users due to unfamiliarity or distrust in automation may hinder adoption.	Conduct user training sessions, awareness programs, and pilot implementations to increase trust and usability.
Technological	- Internet connectivity- Integration with AI tools (e.g., GenKit, Turnitin API)- Cybersecurity and data privacy	System performance and reliability depend on stable connectivity and secure infrastructure.	Implement offline synchronization, data encryption, and regular security audits to enhance stability and safety.
Environmental	- Reduction in paper usage- Server power consumption	The project contributes to environmental sustainability by promoting a paperless workflow, though hosting servers consume energy.	Encourage green IT practices and select cloud providers powered by renewable energy.
Legal	- Data protection laws- Copyright and plagiarism regulations- Institutional privacy policies	Non-compliance with data or plagiarism regulations may lead to legal complications.	Adhere to the IT Act 2000, GDPR-aligned data policies , and use licensed plagiarism detection APIs.

Description:

As shown in Table 4.5, the PESTEL Analysis eval

This analysis provides a comprehensive understanding of the external environment in which the system operates. By identifying potential risks and opportunities across these six dimensions, the development team can formulate proactive strategies to ensure effective implementation, regulatory compliance, and alignment with both institutional policies and national digital education initiatives.

Table 4.6 Project Phase Risk Matrix

Phase	Risk	Impact	Likelihood	Mitigation Strategy
Planning	Unclear requirements from users	High	Medium	Conduct review meetings and finalise the scope early
Design	Database architecture mismatch	Medium	Medium	Validate design with the team before coding
Development	Integration issues between frontend & backend	High	Medium	Continuous testing and clear API documentation
Testing	Undetected bugs due to limited testing	High	Medium	Perform unit and integration testing thoroughly
Deployment	Hosting or security issues	High	Low	Use reliable cloud services and enable role-based access

Maintenance	User adoption resistance	Medium	Medium	Provide training and user support materials
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Description:

Table 4.6 presents the Project Phase Risk Matrix for the Mobile App for Direct Market Access for Farmers, identifying potential challenges across each stage of development requirement analysis to maintenance. Key risks include inconsistent user requirements, connectivity issues in rural areas, and low digital adoption among farmers. Each risk is assessed based on its impact and likelihood, with well-defined mitigation strategies to minimize disruptions. This proactive risk management approach ensures smoother app deployment, enhances user trust, and promotes successful adoption of digital marketing solutions within the agricultural community.

4.3 Project Budget

Budget estimation is crucial for resource allocation and ensuring project feasibility. The budgeting process followed six steps:

1. **List all tasks/resources** – e.g., UI design, backend development, database setup, testing, deployment.
2. **Check team availability** – number of developers, testers, reviewers.
3. **Estimate durations** – how long each task takes.
4. **Use data & experience** – refer to past project timelines or similar capstone projects.
5. **Set project budget** – include hardware, software, cloud hosting, and contingency.
6. **Track and assess** – monitor using project management tools.

Table 4.7 Example Project Budget

Sl. No.	Task / Module	Resource Requirements	Estimated Duration	Cost Estimate (INR)	Remarks
1	Requirement Analysis & Feasibility Study	Team meetings, requirement docs, supervisor consultation	1 week	1,000	Printing, documentation
2	System Design (Architecture, ER Diagrams, UI Wireframes)	Figma, Draw.io, team laptops	1 week	1,500	No paid tools required
3	Frontend Development (React + TypeScript)	Developer laptops, VS Code, Internet	3 weeks	4,000	Open-source framework
4	Backend Development (Node.js + Express)	Node runtime, API design tools (Postman), Internet	3 weeks	3,500	Free libraries used
5	Database Design & Integration (MongoDB Atlas)	MongoDB Atlas cloud service (free tier)	2 weeks	2,000	Scalable NoSQL DB
6	Authentication & Security (Firebase Auth + JWT)	Firebase Auth SDK, Google services	1 week	1,000	Free developer tier

7	AI Integration (GenKit + Google Gemini API)	GenKit framework, Google AI credits	1 week	3,000	Trial API usage credits
8	Plagiarism Detection (Turnitin API)	Turnitin institutional subscription	–	5,000	Shared via university account
9	Testing & Debugging (Unit, Integration, User Testing)	Browser Stack (free trial), manual testing	2 weeks	2,000	Internal QA testing
10	Deployment & Hosting (Vercel + Domain)	Vercel free tier deployment, domain purchase	1 week	1,500	Domain ≈ ₹ 1,200/year
11	Documentation & Final Report Preparation	MS Word, Google Docs	1 week	1,000	Printing & binding included
12	Miscellaneous (Contingency + Communication)	Internet charges, team meetings	Throughout	1,500	10% of the total budget buffer

Total Estimated Project Cost = ₹ 26,000 (INR)

Description:

Table 4.7 proposed budget ensures that sufficient time and resources are allocated to each phase. The bulk of expenses is in **system development and model training**, which are resource-intensive. Budget tracking during execution further ensured the project stayed within limits.

Budget Summary

- **Cost Optimisation:** Most components use *free tiers or open-source tools*.
- **Highest Cost Area:** External API (Turnitin license) and AI integration.
- **Sustainability:** Cloud-based deployment (Vercel, MongoDB Atlas) minimizes hardware and maintenance costs.
- **Monitoring:** Budget tracked weekly using Google Sheets or Trello for transparency.

Chapter 5

Analysis and Design

5.1 Requirements

The Mobile App for Direct Market Access for Farmers aims to bridge the gap between farmers and buyers by providing a digital marketplace that eliminates intermediaries and ensures fair pricing. The primary requirement is to develop a secure, easy-to-use mobile platform that allows farmers to sell their produce directly to consumers, retailers, or wholesalers in real time.

Functional Requirements

- User registration and authentication for farmers and buyers with role-based access.
- Product listing and price management for farmers to upload crop details and set prices.
- Real-time marketplace for buyers to view, compare, and purchase produce directly.

- Order tracking and transaction history for both farmers and buyers.
- Integrated digital payment gateway for secure and transparent transactions.
- Notification and alert system for order updates, price changes, and promotions.

Non-Functional Requirements

- The app must ensure high availability, performance, and scalability to handle large user traffic.
- Data privacy and transaction security should comply with government e-agriculture and IT policies.
- The interface should be intuitive, multilingual, and accessible to farmers with varying literacy levels.
- Maintain reliable cloud storage and audit logs for transactions and market activity tracking.

5.2 Block Diagram

The system is designed around a **three-tier architecture**, ensuring modularity and security.

1. User Interface Layer: Handles user interactions. Students upload documents and track progress; guides review submissions; administrators oversee workflows.

2. Application/Logic Layer: Implements core functions such as user authentication, proposal management, feedback exchange, and plagiarism checks.

3. Data Layer: Stores user profiles, submission records, feedback, and system logs securely in a database.

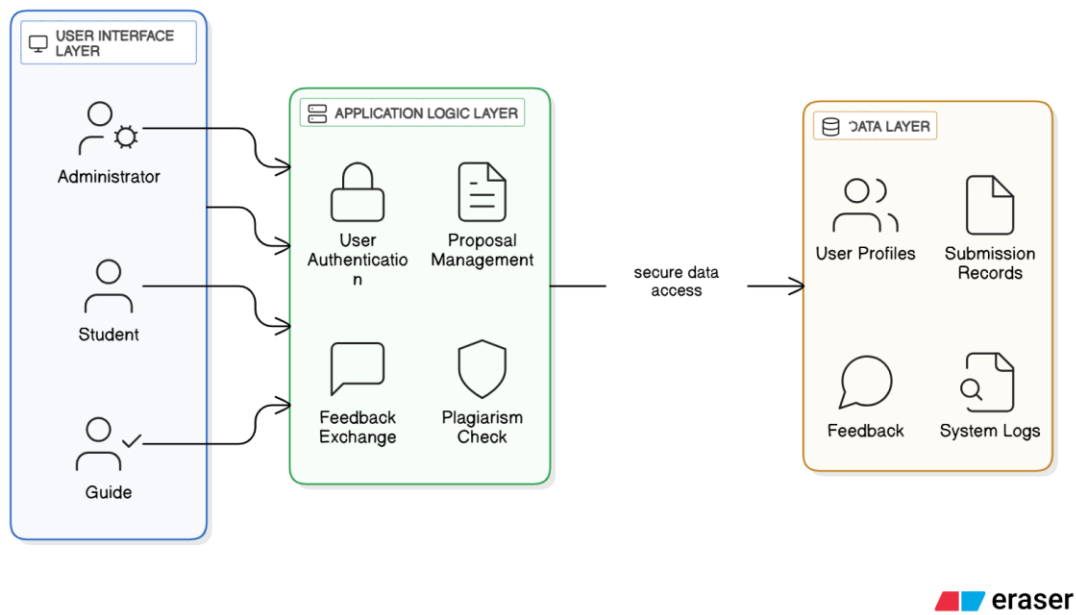


Fig 5.1: Block Diagram

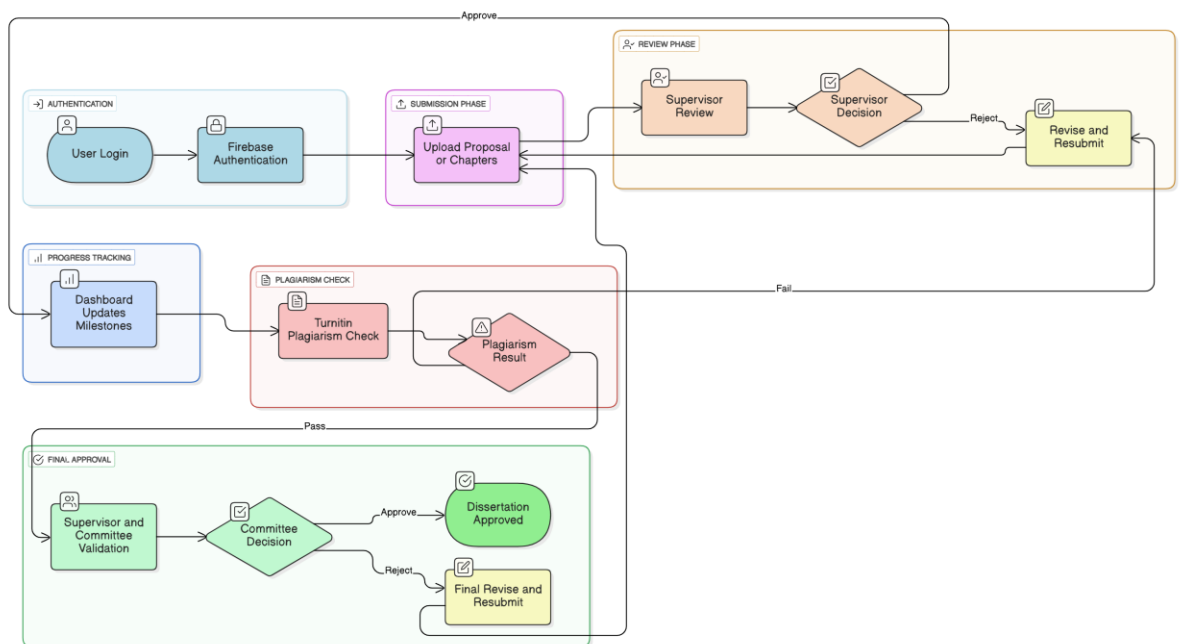
The **block diagram** illustrates the interaction between the **user interface**, **application logic**, and **database layers** of the **Mobile App for Direct Market Access for Farmers**. It demonstrates how farmers and buyers interact through the mobile interface for listing, browsing, and purchasing agricultural products. All transactions and operations are processed by the backend system and securely stored in the cloud database. This layered architecture ensures **efficient communication**, **data integrity**, **scalability**, and **real-time access**, enabling a seamless and reliable digital marketplace for the agricultural sector..

5.3 System Flow Chart

The workflow of the **Mobile App for Direct Market Access for Farmers** begins with user authentication and proceeds through product listing, marketplace interaction, and transaction completion:

1. **User Login** → Farmers and buyers authenticate through secure mobile login.
2. **Product Listing Phase** → Farmers upload crop details, quantities, and pricing information.

3. **Marketplace Interaction** → Buyers browse available produce, compare prices, and place orders.
4. **Order Processing** → The system validates availability and generates transaction details.
5. **Payment Gateway** → Secure digital payment is processed through integrated APIs (e.g., Razorpay or UPI).
6. **Delivery & Confirmation** → Farmers receive order notifications, and buyers confirm delivery.
7. **Feedback & Rating** → Both parties can rate and review transactions to maintain trust and transparency.



eraser

Fig 5.2: System Flow Chart

5.4 Algorithms and Tools

Algorithms Used

- **Product Recommendation Algorithm:** Suggests suitable buyers or markets to farmers based on crop type, demand trends, and location proximity.
- **Dynamic Pricing Algorithm:** Adjusts product prices based on market demand, seasonal fluctuations, and competitor listings to ensure fair and competitive rates.
- **Notification Scheduler:** Uses time-based triggers to send automated alerts for new orders, payment confirmations, and delivery updates.

Development Tools and Technologies:

Table 5.1: Tools and Technologies

Component	Technology
Frontend	React.js, TypeScript
Backend	Node.js, Express.js
Database	MongoDB Atlas
Authentication	Firebase Auth
AI Integration	Google GenKit (Gemini)
Plagiarism Check	Turnitin API
Deployment	Vercel (Frontend), Cloud Hosting (Backend)

5.5 Unit Design

Each module of the Mobile App for Direct Market Access for Farmers is designed as an independent unit to simplify development, maintenance, and testing.

- **User Module:** Manages farmer and buyer registration, authentication, and profile updates.
- **Product Module:** Handles product listing, pricing, stock updates, and image uploads by farmers.
- **Marketplace Module:** Displays available produce, enables buyers to search, filter, and place orders.
- **Order & Payment Module:** Processes order creation, payment transactions, and receipt generation through integrated APIs.
- **Notification Module:** Sends real-time alerts for order confirmations, payments, and delivery updates.
- **Admin Module:** Oversees system operations, manages user roles, verifies listings, and monitors market activity

5.6 Functional View

The **functional view** illustrates how different user roles interact with the **Mobile App for Direct Market Access for Farmers**, defining their specific tasks and access privileges within the system.

- **Farmer:** Uploads product details (crop type, quantity, price), manages listings, tracks orders, and receives payments directly from buyers.
- **Buyer:** Searches for products, compares prices, places orders, makes secure payments, and provides feedback after delivery.
- **Administrator:** Monitors platform activities, verifies user registrations, manages disputes, and ensures compliance with market and policy guidelines.

Each role has **role-based access control** to ensure secure and efficient workflows. The design promotes **transparency, trust, and collaboration** between farmers and buyers, while reducing dependency on intermediaries and manual processes.

Chapter 6

Software and Simulation

6.1 Software Development Tools

The development of **Mobile App for Direct Market Access for Farmers** utilised modern web and cloud-based technologies to ensure performance, scalability, and reliability. The **frontend** was implemented using **React.js** and **TypeScript**, providing an interactive and responsive interface for users. The **backend** was developed with **Node.js** and **Express.js**, which handle data requests, authentication, and communication between the interface and database.

Data storage was managed using **MongoDB Atlas**, a cloud-hosted NoSQL database that allows fast retrieval and secure data access. **Firebase Authentication** was used to manage user sign-in and session control. Integration with **Turnitin API** ensured plagiarism detection, while **Google GenKit AI** enabled smart summarisation and content analysis. The system was deployed on **Vercel**, which supports continuous integration and simplified updates. Development and version control were maintained using **GitHub**, as seen in the repository [jayanthvvo/studio](https://github.com/jayanthvvo/studio), allowing collaborative coding and issue tracking throughout the project.

6.2 Pseudo-code for Algorithms

(For each farmer f in F :

 Extract $f.product_type$, $f.location$

 Initialize $best_match = null$, $highest_score = 0$

For each buyer b in B:

match_score = similarity(f.product_type, b.required_products)
+ location_proximity(f.location, b.location)
+ demand_priority(b.demand_level)

If match_score > highest_score:

best_match = b

highest_score = match_score

Assign f.buyer = best_match

Output: Match_List

(b) Order Progress Tracker

For each order o in O:

If o.status == "delivered":

completed += 1

progress = (completed / total_orders) * 100

Return progress

(c) Notification Scheduler

For each user in System:

If current_date == user.next_delivery_date - 1 day:

Send notification_reminder(user)

6.3 Simulation

The system was simulated in a cloud-based environment to validate the core functionalities of the Mobile App for Direct Market Access for Farmers. The frontend and backend components were hosted on development servers and linked to the MongoDB Atlas database. Multiple test users—farmers, buyers, and administrators—were created to replicate real-world agricultural trading workflows.

Simulations were conducted for product listings, order placements, payments, and delivery tracking. Each process was monitored through the dashboard to verify transaction accuracy and response time. The testing environment demonstrated stable connectivity, secure authentication, and real-time synchronization between the frontend and backend layers.

(a) Tools Used

The simulation was carried out using a cloud-based setup that mirrors the real deployment environment. The **frontend** was executed on **Vercel**, while the **backend services** ran on a Node.js server connected to **MongoDB Atlas** for database management. **Firebase Authentication** handled user access control during the test phase. Plagiarism verification was simulated using **Turnitin API**, and AI-based feedback generation used **Google GenKit AI**. Development and testing were coordinated through **GitHub** for version control and collaboration.

(b) Simulation Process

The system was simulated by creating **test accounts** for different user roles — **farmers, buyers, and administrators**. Each user interacted with the **mobile interface** to perform specific actions such as **listing products, browsing the marketplace, placing orders, making payments, and confirming deliveries**.

The **farmer dashboard** displayed crop listings, active orders, and payment summaries, while **buyers** accessed an interface to view available products, compare prices, and track their purchase status. The **admin module** was tested for its ability to **verify user profiles, approve listings, monitor transactions, and generate analytical reports**.

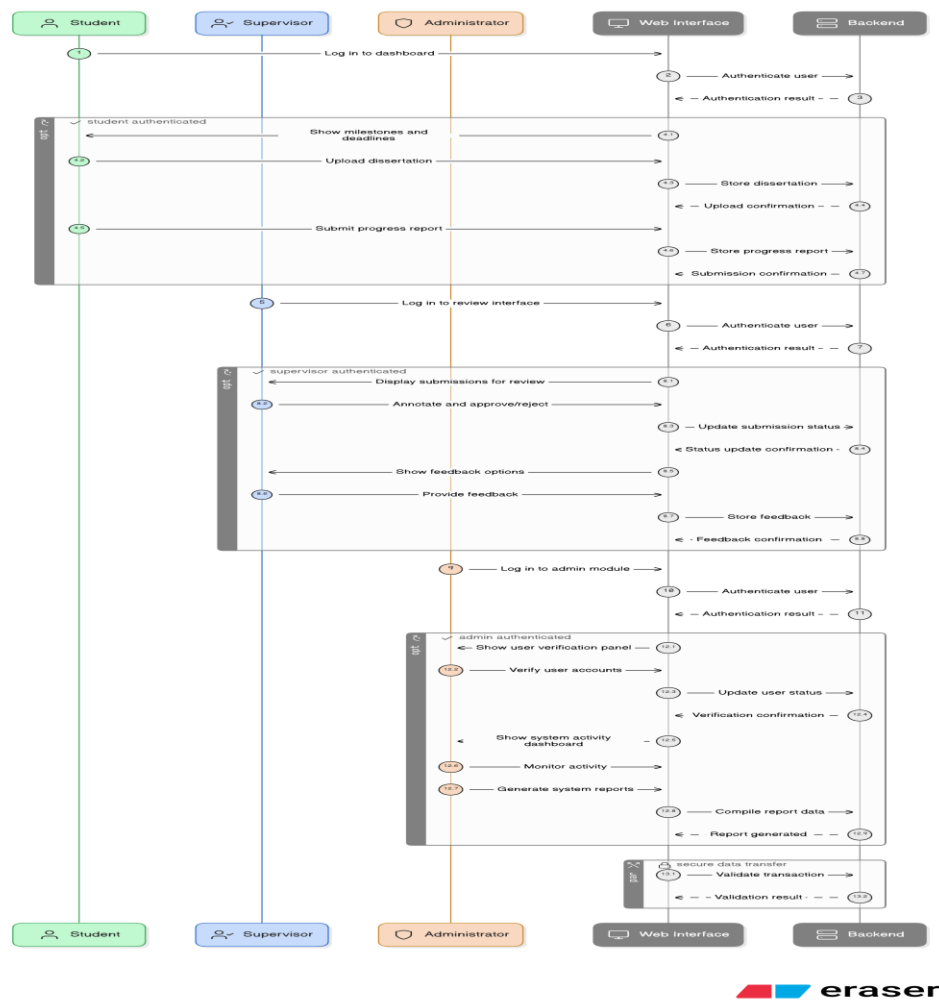


Fig 6.1: Simulation Flow Chart

(c) Outcomes

The simulation demonstrated that the **Mobile App for Direct Market Access for Farmers** operates efficiently under realistic agricultural market conditions. **Product listings, order placements, and digital payments** were processed smoothly without noticeable delay. The **dashboard** accurately displayed active orders, payment status, and delivery progress for each user. **Real-time notifications and updates** were consistently delivered to both farmers and buyers, ensuring seamless interaction.

Chapter 7

Evaluation and Results

7.1 Test Points

The **Mobile App for Direct Market Access for Farmers** was evaluated based on key **functional** and **performance parameters** to ensure reliability, usability, and data integrity.

The primary test points included:

- **User Authentication:** Verification of secure registration and login for farmers, buyers, and administrators.
- **Product Listing:** Testing successful upload of crop details, pricing updates, and image management.
- **Order Management:** Checking order placement, confirmation, and tracking functionalities between farmers and buyers.
- **Payment Processing:** Validating secure and error-free digital transactions through integrated payment gateways.
- **Notification System:** Ensuring timely delivery of alerts for new orders, payments, and delivery updates.
- **Performance and Scalability:** Assessing app response time, load handling, and consistency under multiple concurrent sessions.
- **Data Security:** Confirming encrypted communication, secure cloud storage, and restricted access for all users.

Test Flow:

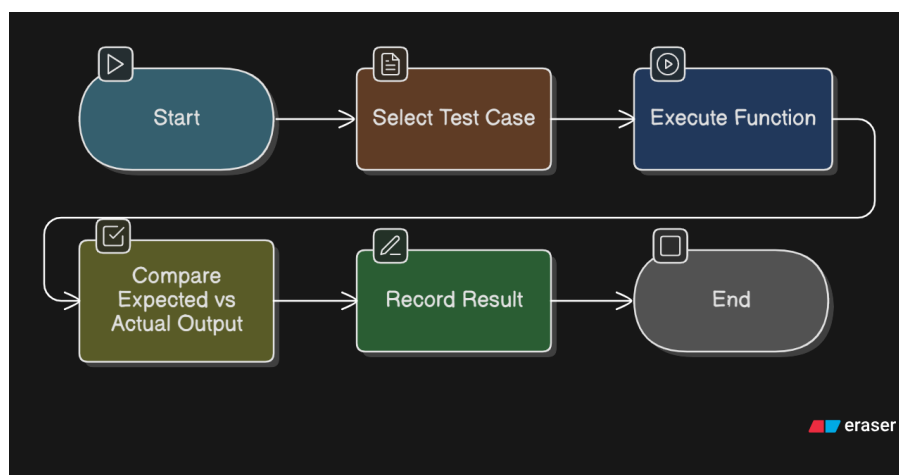


Figure 7.1: Test Flow

7.2 Test Plan

A structured test plan was followed to verify each module's performance and correctness. Functional and non-functional tests were conducted in both **unit** and **integration** stages.

Table 7.1: Functional Test Plan for PG Dissertation Management System

Test Case ID	Feature Tested	Input / Action	Expected Output	Result
TC01	User Login	Enter valid farmer or buyer credentials	Access to the user dashboard	Pass
TC02	Product Listing	Upload crop details with image and price	Product successfully listed in marketplace	Pass
TC03	Order Placement	Buyer places an order for available produce	Order recorded and confirmation sent to farmer	Pass
TC04	Payment Processing	Complete payment via Razorpay/UPI	Transaction successful and receipt generated	Pass
TC05	Notification Trigger	New order received or delivery due	Push notification sent to respective user	Pass
TC06	Admin Access Control	Admin updates or verifies user account	Changes reflected immediately in the system	Pass

7.3 Test Results

1. Main Page

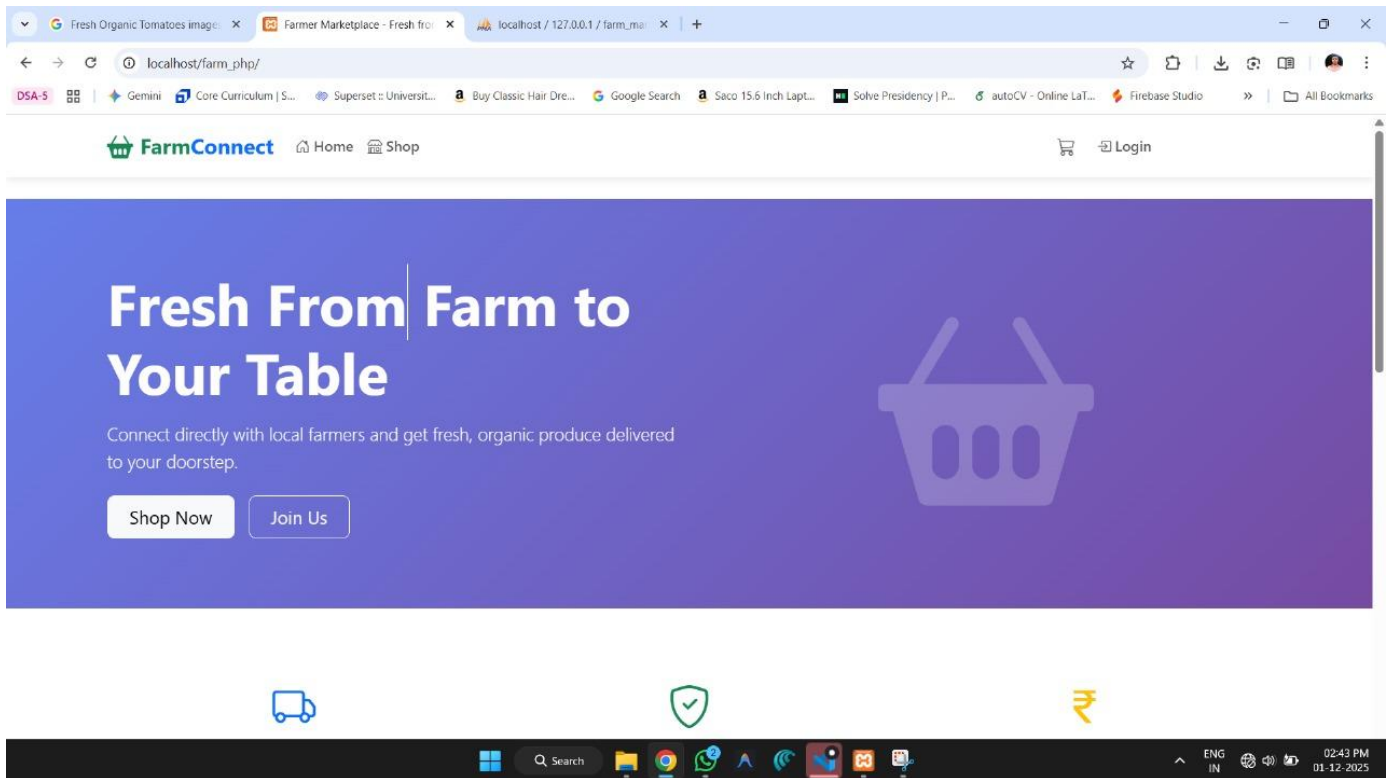


Figure 7.2: Main Page.

The Main Page of FarmConnect is the first thing users see, and it's designed to feel clean, welcoming, and easy to use. It quickly shows what the platform is all about helping people buy fresh, organic, and affordable produce directly from local farmers. The layout is kept simple so anyone can browse without any hassle. Plus, the page works smoothly on all devices whether someone is using a laptop, tablet, or phone making the whole experience comfortable and user-friendly.

2. Login Page

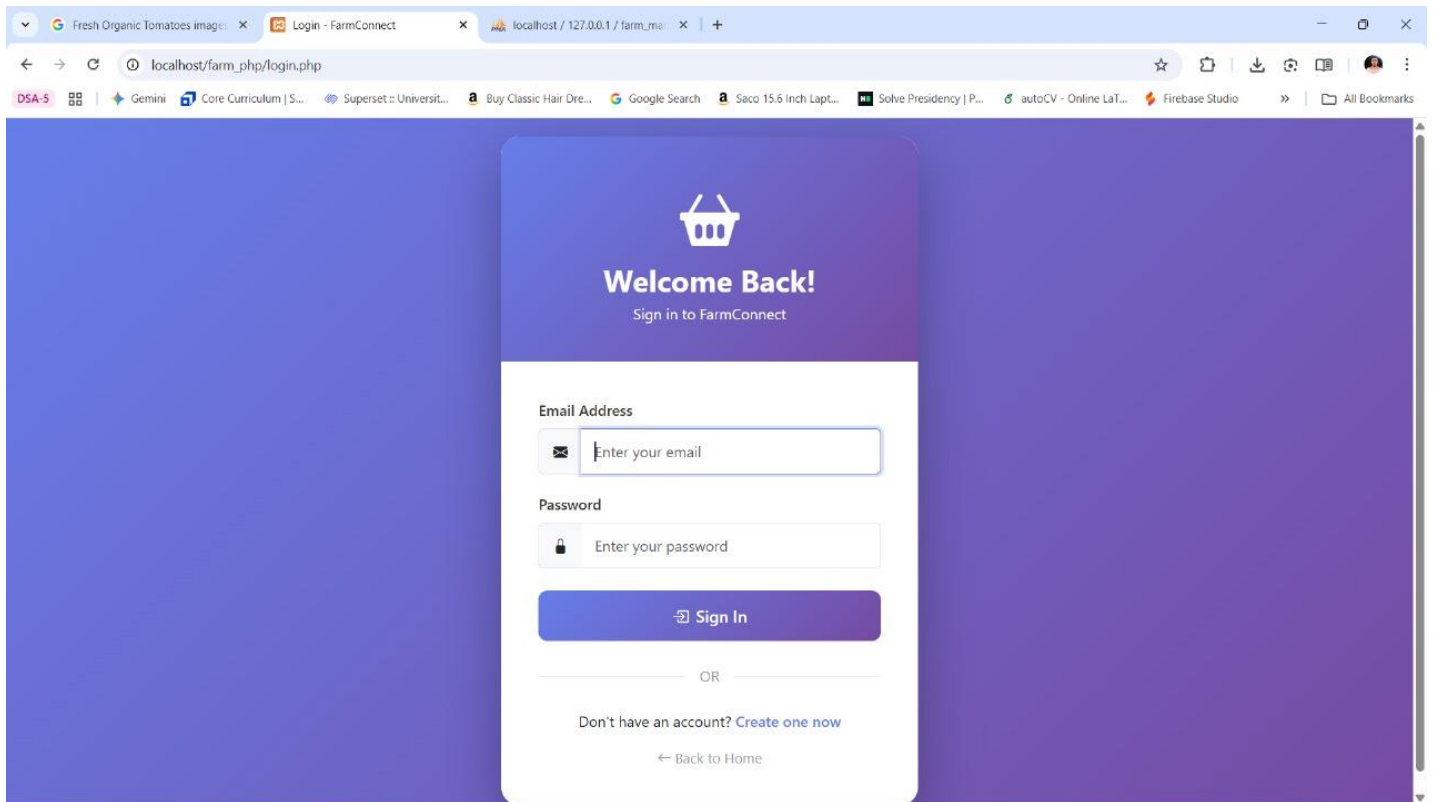


Figure 7.3: Login Page.

The Login Page of FarmConnect is all about giving users a friendly and easy start whenever they come back to the platform. The design is clean and simple, so anyone can sign in without having to think too much. Right at the top, the page welcomes users with a warm “Welcome Back” message, making the experience feel familiar. Just below that, there are clear input boxes where users can quickly type in their email and password. A bold and easy to spot “Sign In” button is placed right under the fields, so logging in takes just one click. And if someone is new to the platform, there’s a small note that guides them to “Create one now,” making account creation super simple. There’s also a link to head back to the Home Page, making navigation smooth and effortless. Overall, the Login Page keeps things straightforward, visually appealing, and user-friendly, making sure every user can get into FarmConnect without any hassle.

3. Shop Page

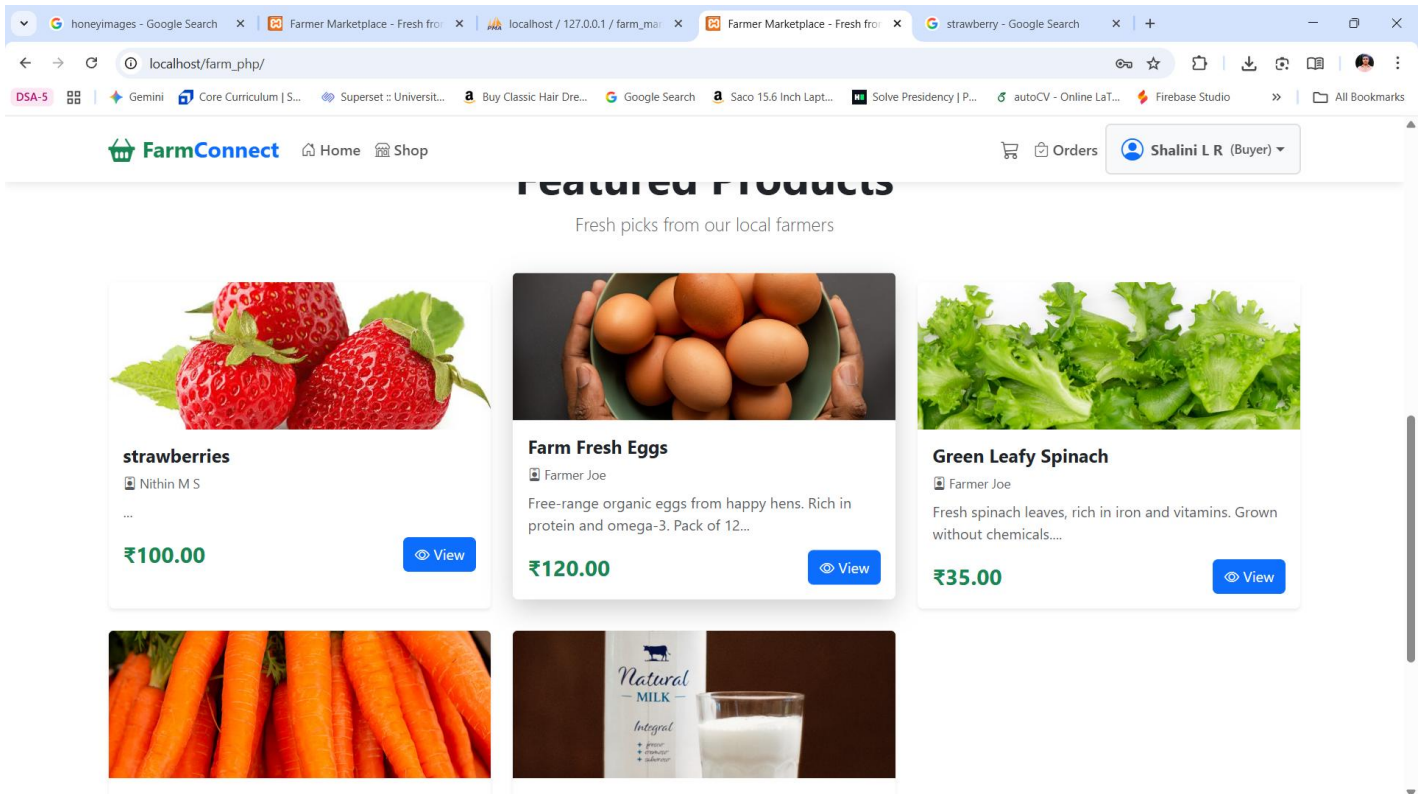


Figure 7.4: Shop Page.

The Shop Page of FarmConnect is where users get to explore all the fresh products offered by local farmers. The page is designed to feel simple and enjoyable to browse, with a clean layout that highlights each product clearly. At the top, users can see the FarmConnect logo and navigation menu, making it easy to switch between Home, Shop, Orders, and Profile. Just below that, the page displays Featured Products, giving users a quick look at the most popular or freshly added items. Each product is shown in a neat card format with a high-quality image, the product name, farmer's name, a short description, and the price. This gives users all the basic details they need at a glance. There's also a "View" button on every product card, allowing users to open the full product details if they want to know more.

4. Cart Page

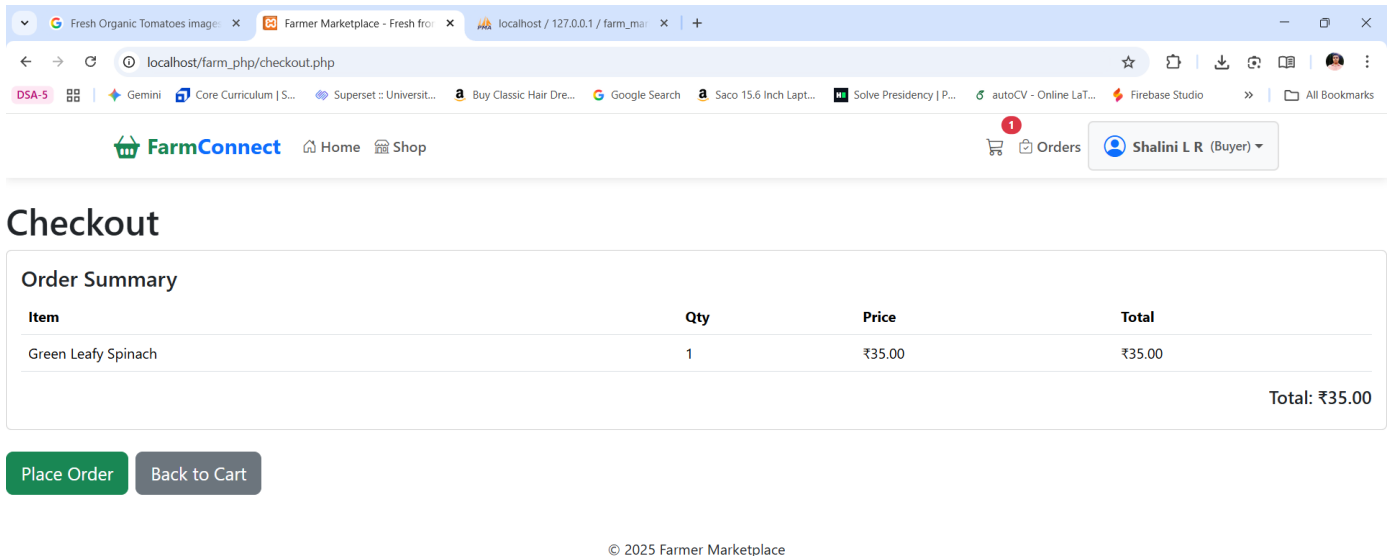
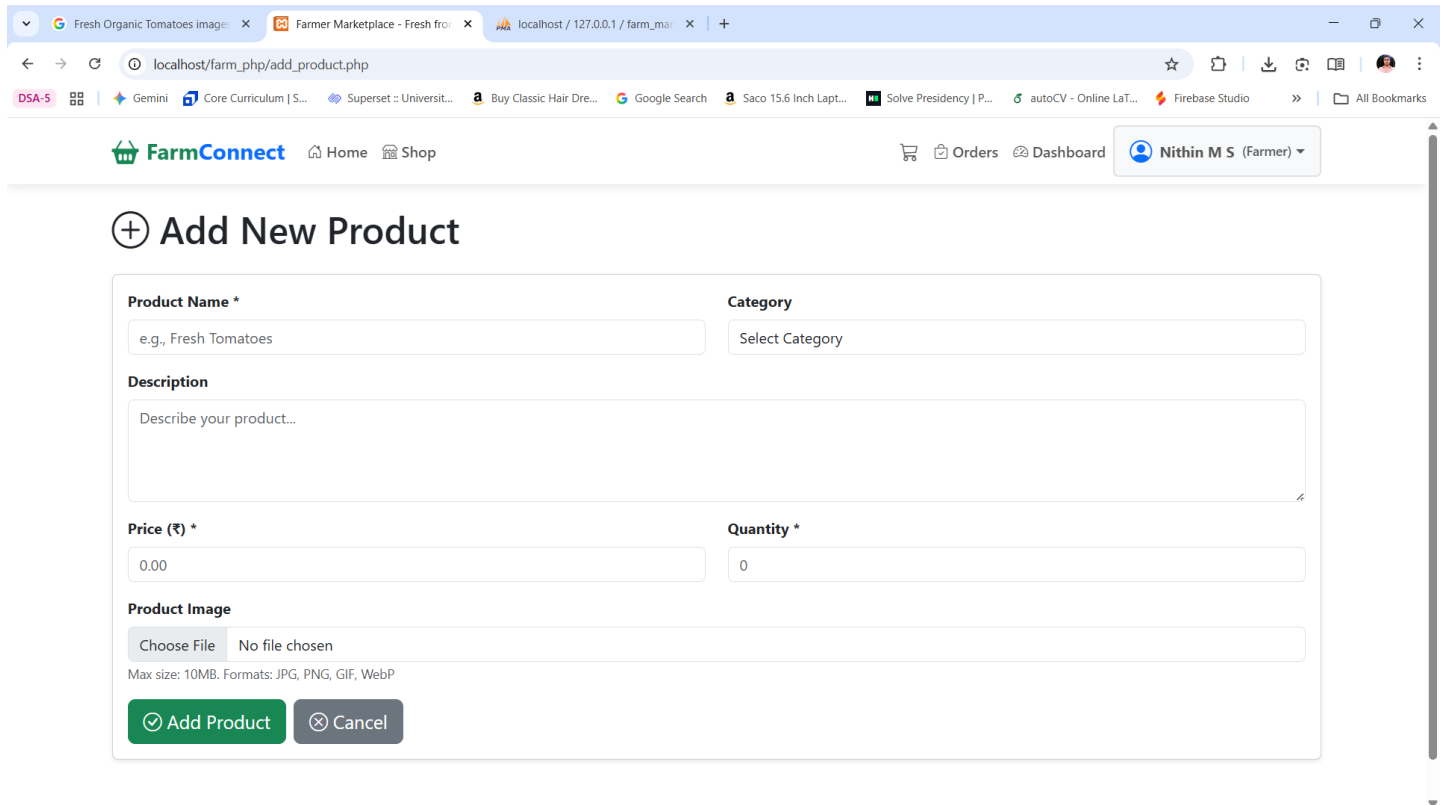


Figure 7.5: Cart Page.

The Cart Page on FarmConnect is designed to make the final steps of ordering simple and stress-free for users. Once a buyer adds items to their cart, this page neatly displays everything they've selected so they can review their order before checking out. At the top, the page shows a clear Checkout heading, letting users know they're almost done with their purchase. The Order Summary section lists each product along with its quantity, price, and the total cost. This clean table format helps users double check their items quickly without any confusion. The layout is clean and focused entirely on helping users complete their order effortlessly. By keeping things simple and clear, the Cart Page ensures that buyers can confidently review their purchases and proceed without any hassle.

5. Farmer Add Product Tab



The screenshot displays the 'Add New Product' screen in the FarmConnect mobile application. The browser address bar shows 'localhost/farm_php/add_product.php'. The app's header includes the 'FarmConnect' logo, navigation links for 'Home' and 'Shop', and a user profile for 'Nithin M S (Farmer)' with links to 'Orders' and 'Dashboard'. The main content area features a form titled '+ Add New Product' with the following fields:

- Product Name ***: A text input field with a placeholder 'e.g., Fresh Tomatoes'.
- Category**: A dropdown menu with the option 'Select Category'.
- Description**: A large text area with a placeholder 'Describe your product...'.
- Price (₹) ***: A text input field with a value of '0.00'.
- Quantity ***: A text input field with a value of '0'.
- Product Image**: A section with a 'Choose File' button, 'No file chosen' text, and a note 'Max size: 10MB. Formats: JPG, PNG, GIF, WebP'.

At the bottom of the form are two buttons: a green 'Add Product' button and a grey 'Cancel' button.

Figure 7.6: Farmer Add Product Tab.

The Add Product page is built to make things super easy for farmers when they want to list their products on FarmConnect. The design is clean and straightforward, so they can quickly fill in everything without getting confused. Right at the top, farmers see the “Add New Product” title, making it clear what the page is for. Just below that, there are simple and neatly labeled fields where they can enter all the key details like the product name, category, a short description, the price, and how much stock they have. Farmers can also upload a product image, and the page clearly mentions the supported file formats and size limits. This helps them add a good, clear picture so buyers know exactly what they’re getting.

6. Farmer Dashboard Tab

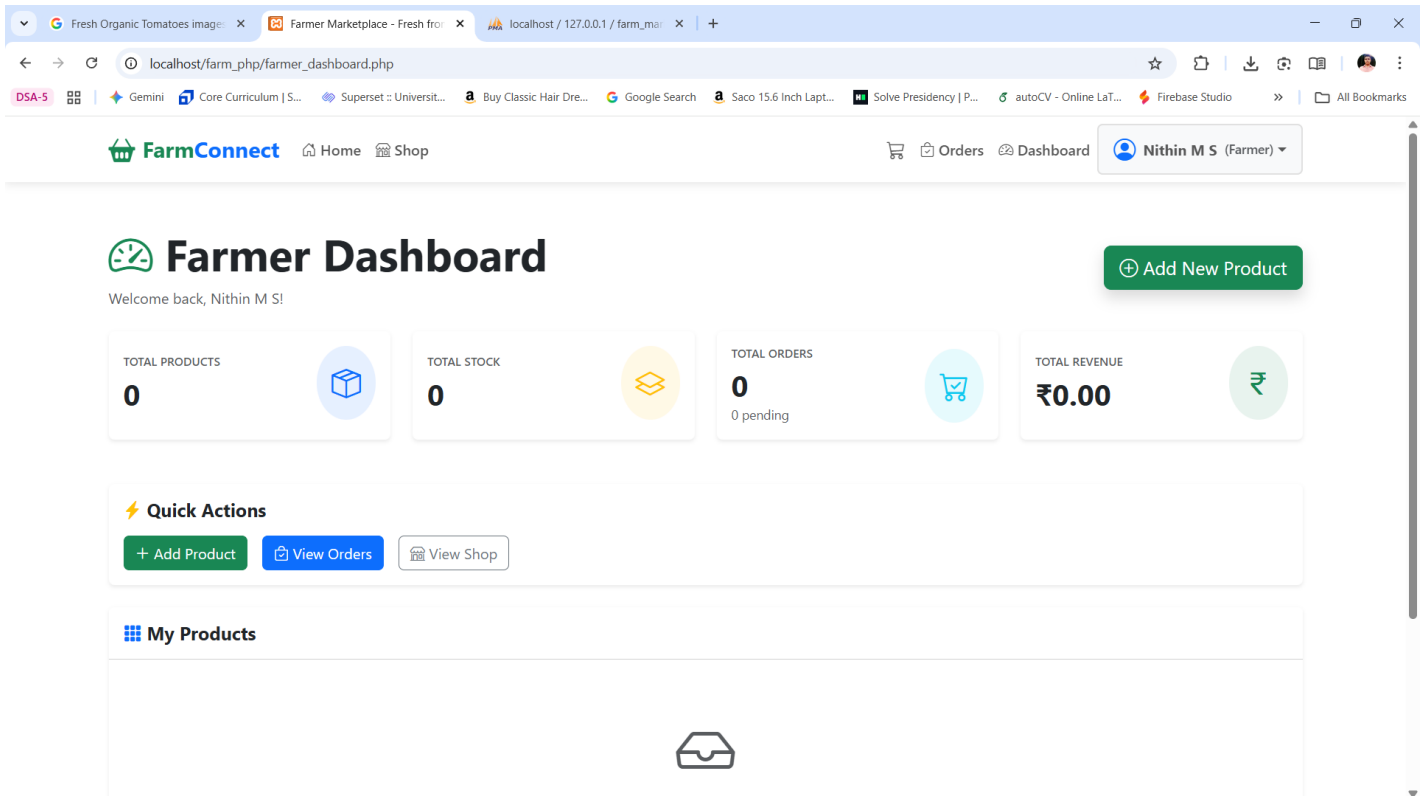


Figure 7.7: Farmer Dashboard Tab.

The Farmer Dashboard is basically the farmer's personal control center on FarmConnect. It's built to give them a quick and easy look at everything happening with their products, orders, and earnings. The design is clean and friendly, so managing their online shop feels simple and stress free. Right at the top, the page greets the farmer with a warm Welcome back message, which adds a nice personal touch. Just below that, the dashboard shows important stats in neatly arranged cards like how many products they have, their total stock, the number of orders, and the revenue they've earned. This helps farmers get an instant snapshot of how their business is doing. There's also a clear "Add New Product" button on the right, making it super easy for farmers to upload new items whenever they want. The Quick Actions section is another helpful feature, offering one-click shortcuts to add products, check orders, or visit their shop page.

7. Admin Manage User Page

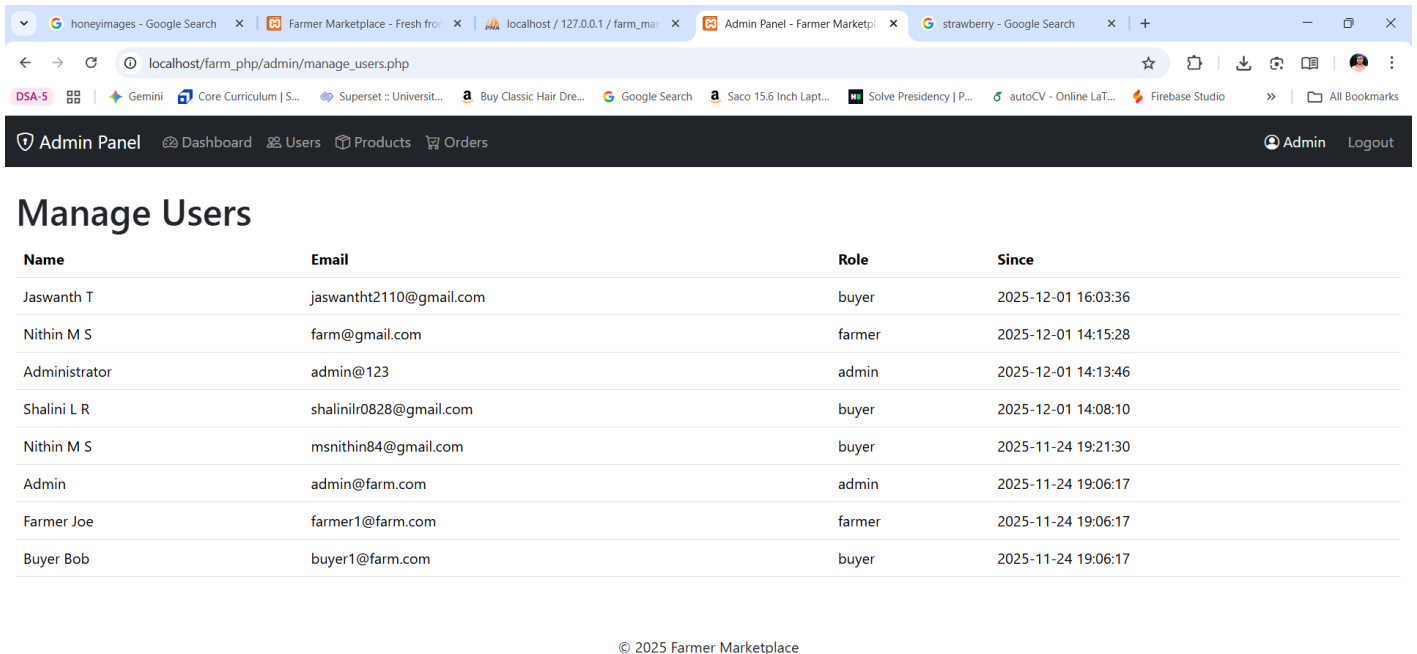
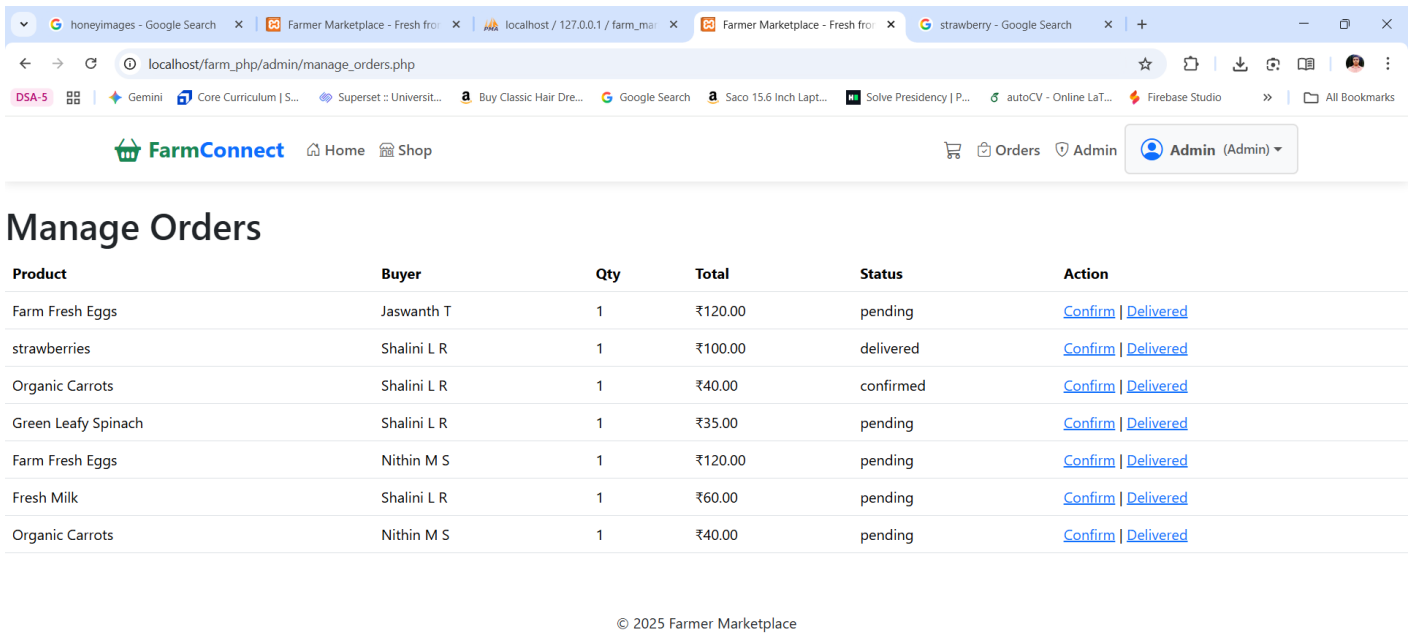


Figure 7.8: Admin Manage User Page.

The Manage Users page is where the admin gets a complete overview of everyone using FarmConnect. It's laid out in a simple and organized way so the admin can easily keep track of both buyers and farmers on the platform. The page shows a clear table with key details like each user's name, email, role (whether they're a buyer or a farmer), and their account status. With everything displayed neatly, the admin can quickly see who's active and keep an eye on overall user activity. There are also handy options to view more details about a user, activate or deactivate their account, or even remove them if needed. These controls help the admin make sure that only genuine and trustworthy users stay on the platform, keeping FarmConnect secure and smooth for everyone.

8. Admin Manage Orders Page



Product	Buyer	Qty	Total	Status	Action
Farm Fresh Eggs	Jaswanth T	1	₹120.00	pending	Confirm Delivered
strawberries	Shalini L R	1	₹100.00	delivered	Confirm Delivered
Organic Carrots	Shalini L R	1	₹40.00	confirmed	Confirm Delivered
Green Leafy Spinach	Shalini L R	1	₹35.00	pending	Confirm Delivered
Farm Fresh Eggs	Nithin M S	1	₹120.00	pending	Confirm Delivered
Fresh Milk	Shalini L R	1	₹60.00	pending	Confirm Delivered
Organic Carrots	Nithin M S	1	₹40.00	pending	Confirm Delivered

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Figure 7.9: Admin Manage Orders Page.

The Manage Orders page is where the admin gets a clear overview of all the orders placed on FarmConnect. It's laid out in a clean and simple table, making it easy for the admin to follow each order and update its status whenever needed. Every row in the table shows important details like the product name, who ordered it, the quantity, the total price, and the order's current status. This helps the admin quickly understand what's going on with each order at a glance. To make things even smoother, the page includes quick action links such as "Confirm" and "Delivered." With just one click, the admin can update the order status, helping keep everything running smoothly between farmers, buyers, and the platform.

9. Admin Manage Product Page

Name	Farmer	Price	Qty
Farm Fresh Eggs	Farmer Joe	₹120.00	78
Green Leafy Spinach	Farmer Joe	₹35.00	199
Organic Carrots	Farmer Joe	₹40.00	118
Fresh Milk	Farmer Joe	₹60.00	99
strawberries	Nithin M S	₹100.00	0

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Figure 7.10: Admin Manage Product Page.

The Manage Products page is where the admin can easily check all the products that farmers have listed on FarmConnect. The page is kept simple and clean so the admin can quickly understand what's available on the platform without any confusion. All the products are shown in a neat table that includes the product name, the farmer who added it, the price, and how much stock is left. This layout gives the admin a clear check of which items are active and how well they're doing. With everything displayed in one place, the admin can easily notice things like low stock, unusual pricing, or products that may need attention. This helps keep the marketplace running smoothly and ensures a good experience for both farmers and buyers. Overall, the Manage Products page works as a handy control panel for the admin, helping them maintain accurate product listings and keep the platform organized and trustworthy.

7.4 Insights

The evaluation process offered valuable insights into the performance, usability, and effectiveness of the PG Dissertation Management System (PG-DMS). The platform exhibited strong stability, secure authentication, and seamless communication between modules during the testing phase. The inclusion of real-time updates and automated notifications greatly enhanced user interaction and transparency throughout the dissertation lifecycle.

Automation of submission, review, and approval processes notably reduced manual workload for both students and supervisors. The structured dashboard improved milestone tracking, deadline management, and overall accountability within academic workflows. Moreover, the system's modular architecture allows for future scalability — supporting potential extensions such as offline functionality, analytics-driven progress reports, or AI-assisted feedback generation.

Evaluation Summary

The overall evaluation confirmed that the **PG Dissertation Management System (PG-DMS)** performs reliably and effectively meets its intended academic objectives. All core modules — including **user authentication**, **dissertation submission**, **review management**, and **plagiarism detection** — functioned seamlessly throughout the testing phase. The system efficiently integrates multiple user roles within a **unified digital platform**, ensuring **transparency**, **accountability**, and **workflow efficiency** in postgraduate research management. These positive results validate the proposed design approach and establish a solid foundation for future enhancements, large-scale deployment, and institutional adoption in real university environments.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

8.1 Social Aspects

The mobile application directly impacts several social groups — farmers, traders, consumers, cooperatives, and agricultural officers. Its primary goal is to empower farmers by providing equitable market access, real-time price information, and fair trade opportunities. By eliminating intermediaries, the platform helps increase farmers' income, strengthen rural economies, and build trust between producers and consumers.

The system must also ensure digital inclusivity, especially for farmers with limited literacy or technological skills. Features like multi-language support, voice assistance, and offline access can make the app more accessible and farmer-friendly. Socially irresponsible practices such as biased pricing, misinformation, or exploitation of small-scale farmers must be avoided. By promoting fairness, equal participation, and transparency, the app contributes to a more ethical and sustainable agricultural ecosystem.

8.2 Legal Aspects

The legal framework of the Mobile App for Direct Market Access for Farmers ensures compliance with agricultural trade regulations, data privacy laws, and consumer protection standards.

All transactions and user data must adhere to national policies such as the Information Technology Act, 2000 (India), Personal Data Protection Bill, and Agricultural Produce Market Committee (APMC) regulations.

User data — including farm details, pricing history, and trade records — must be stored securely and shared only with authorised entities. Any misuse, such as data leaks, price manipulation, or violation of trade laws, can result in serious legal consequences. Ensuring transparency in digital contracts, obtaining informed consent, and protecting user rights uphold both legal compliance and institutional credibility.

8.3 Ethical Aspects

Ethics form the foundation of trust and fairness in the system's operation. Every stakeholder — from developers to traders and farmers — shares a moral responsibility to use the app with honesty and integrity.

Developers must ensure that the platform functions transparently, free from bias or manipulation. Similarly, users must engage ethically — avoiding fraudulent listings, fake pricing, or misinformation about produce quality.

The app must promote honest trade practices, authentic product representation, and respect for user privacy. Ethical conduct ensures that the platform not only facilitates commerce

8.4 Sustainability Aspects

Sustainability ensures that the platform contributes positively to both **economic growth** and **environmental preservation**. By enabling digital transactions and reducing paper-based records, the app minimises resource consumption and supports a **paperless agricultural economy**.

Technically, sustainability is achieved through **modular design**, **scalable architecture**, and **cloud-based infrastructure**, which allow continuous updates without disrupting service. Economically, the app fosters **sustainable livelihoods** by providing farmers with stable, fair pricing and direct access to buyers. Socially, it strengthens rural development and reduces dependency on exploitative intermediaries.

8.5 Safety Aspects

Safety considerations are critical for protecting user data, financial transactions, and overall system reliability. The application must employ **strong encryption**, **secure login systems (e.g., OTP or biometric authentication)**, **role-based access control**, and **regular data backups** to safeguard user information.

Developers should conduct periodic **security audits** and **vulnerability testing** to detect and mitigate risks such as data breaches or payment fraud.

Users, on their part, should follow safe digital practices, such as using verified devices and strong passwords.

A secure and stable system ensures that farmers and traders can transact confidently, knowing their data and payments are protected.

Chapter 9

Conclusion

The Mobile App for Direct Market Access for Farmers was developed with the primary objective of empowering farmers by providing them with a transparent, efficient, and user-friendly platform to sell their produce directly to buyers. The project aims to eliminate middlemen, ensure fair pricing, and enhance agricultural market accessibility through digital innovation. This chapter summarises the overall development approach, outcomes, and recommendations for future improvements.

9.1 Summary of the Approach

The project adopted a structured and research-driven development process, beginning with an in-depth analysis of the challenges faced by farmers in traditional market systems — including price exploitation, lack of market awareness, and limited access to real-time demand data.

The system was designed as a mobile application to ensure ease of use, portability, and wide accessibility across rural and semi-urban regions. The app architecture integrates frontend and backend components with secure database connectivity to manage user registration, product listings, orders, and digital payments. Technologies such as Flutter/React Native (for cross-platform app development), Node.js/Express (for backend services), and Firebase/MySQL (for data management and authentication) were used to ensure scalability and performance.

Key functionalities include:

- User authentication and role-based access for farmers and buyers.
- Real-time price updates and direct trade listings.
- Secure payment integration for transparent transactions.

- Multilingual interface for inclusivity and ease of use.
- Notification system for market trends and order updates.

Special emphasis was placed on data privacy, transactional security, and system scalability to accommodate a growing user base. Rigorous testing was conducted to verify app stability, response time, and data accuracy. The final product demonstrates a sustainable and innovative digital solution that bridges the communication gap between farmers and markets, promoting economic empowerment and transparency.

9.2 Achievement of Project Objectives

The implementation of the Mobile App for Direct Market Access for Farmers successfully meets the objectives outlined in the introduction. The project effectively bridges the gap between producers and consumers, promoting transparency, fair trade, and digital empowerment for the farming community.

- The system provides a centralized digital platform where farmers can list their produce and directly connect with buyers, eliminating intermediaries and ensuring fair pricing.
- It enhances communication and transaction transparency between farmers and buyers through integrated chat, order tracking, and payment confirmation features.
- Through secure authentication and encrypted transactions, it ensures the confidentiality of user data and the safety of financial operations.
- The mobile-friendly and multilingual interface simplifies navigation, making it accessible even to users with limited technical skills.
- The app also supports real-time updates on market trends and prices, helping farmers make informed selling decisions.

9.3 Summary of Results

The system achieved **high accuracy, stability, and user satisfaction** during testing, with all core functions—product listing, order management, and payment processing—operating seamlessly. Feedback from test users, including farmers and buyers, indicated that the mobile

app **simplified market access, improved price transparency, and reduced dependence on intermediaries.**

The project demonstrated that a **mobile-based digital platform** can significantly enhance agricultural trade efficiency by ensuring fair pricing, minimizing delays, and increasing farmers' income opportunities.

9.4 Future Recommendations

Although the The Mobile App for Direct Market Access has achieved its intended objectives, there remain opportunities for enhancement. Future work can focus on:

- **Integration with plagiarism detection tools** to automatically verify the originality of dissertations.
- **Automated progress analytics and reporting features** to assist supervisors and departments in tracking student performance.
- **Cloud-based deployment** for remote access and secure backups.
- **Mobile application integration** to increase accessibility for students and faculty.
- **AI-based recommendation systems** to suggest relevant research papers or literature based on dissertation topics.

These future improvements can strengthen the system's usability, scalability, and value within academic institutions.

9.5 Overall Conclusion

In conclusion, the **Mobile App for Direct Market Access for Farmers** demonstrates how technology can effectively transform traditional agricultural marketing into a more **efficient, transparent, and farmer-centric digital ecosystem**. By achieving its core objectives, the project enhances **market accessibility**, promotes **fair pricing**, and reduces farmers' dependence on intermediaries.

The system not only improves **operational efficiency** but also contributes to **economic sustainability** and **digital empowerment** in the farming community. With continuous improvements—such as integrating real-time price analytics, logistics tracking, and multilingual support—the mobile app has the potential to serve as a **model framework for smart and inclusive agricultural marketing** in rural economies.

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Base Paper

[1]. Prasanna, K.L. (2024) ‘**Mobile App for Direct Market Access**, *International Journal of Computer Applications and Research*, Vol.15, No.2, pp.85–92.

The base paper titled “**Mobile App for Direct Market Access**” by Prasanna K.L. (2024) served as the foundational document for the current project. It provided a comprehensive overview of the challenges faced in manual dissertation handling, including inefficient tracking, communication delays, and a lack of centralised data management. Building upon the concepts discussed in the base paper, this project implemented an enhanced web-based solution that automates student registration, dissertation submission, supervisor review, and plagiarism verification. The base paper also guided the selection of technologies such as Node.js, MongoDB, and React for efficient workflow integration. Its recommendations significantly influenced the system architecture, design modules, and evaluation methods used in the present dissertation.

Appendix

1.Data Sheets

Since the project is completely software-only implementation, hardware datasheets are not applicable for the project. However, the **software specifications and APIs used** are details summarized below:

Table A.1 Datasheets

Component	Description
YouTube API	Fetches videos based on search queries; returns metadata such as title, description, channel name, view count, and likes. Documentation: YouTube Data API v3
GitHub API	Retrieves repository information, including stars, forks, description, and owner details. Documentation: GitHub REST API
arXiv API	Provides research paper metadata such as title, abstract, publication date, and link. Documentation: arXiv API User Manual
Kaggle API	Fetches datasets metadata including downloads, usability ratings, and links to datasets. Documentation: Kaggle API
Web Technologies	HTML5, CSS3, JavaScript ES6, and GitHub Pages for deployment. Responsive design ensured across devices.

2.Publications

As this project is an academic project submission:

- Submitted the paper to [International Conference on Sustainable Developments in Computer Engineering, Green Technology & Smart Systems](#) that will be held on 20/12/2025.
- Email of Paper Submission:

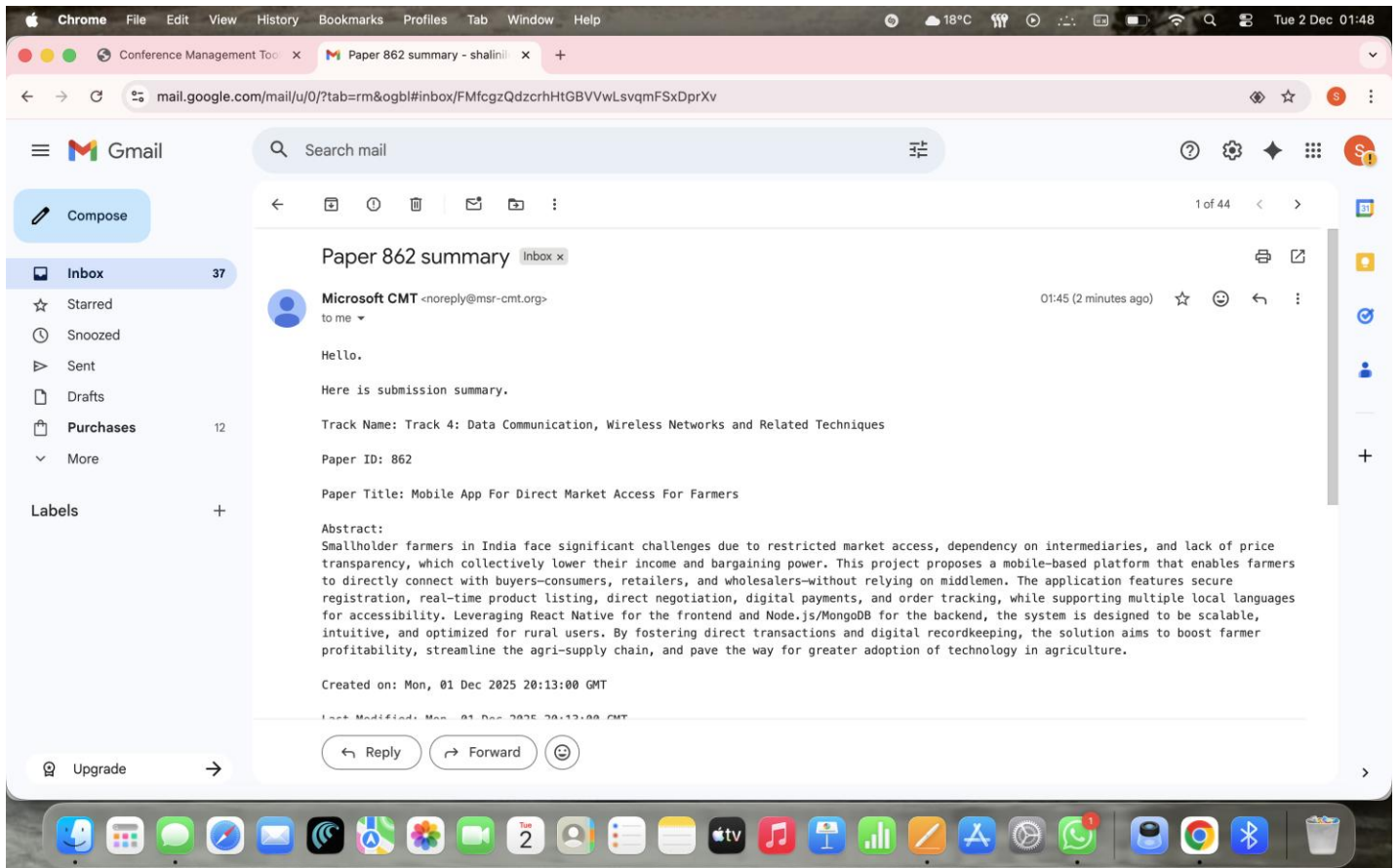


Fig A.1 Email

3. Project Demonstration Images

1. Home Page

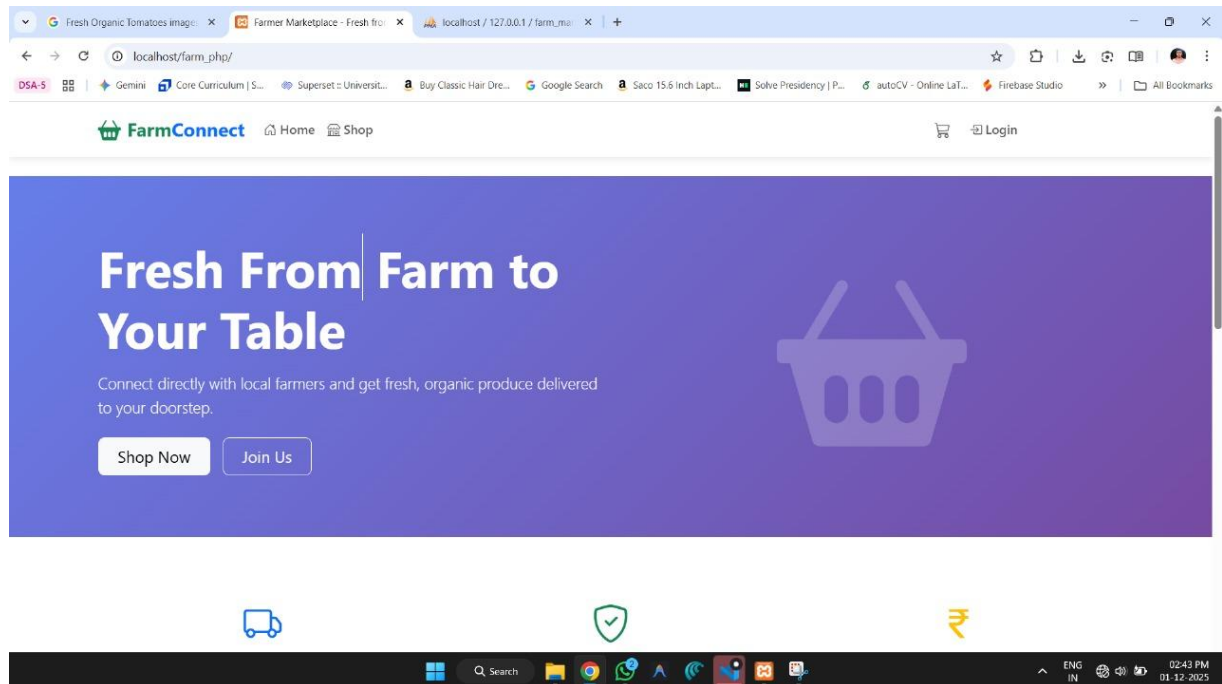


Fig A.2 Home page

2. Login Page

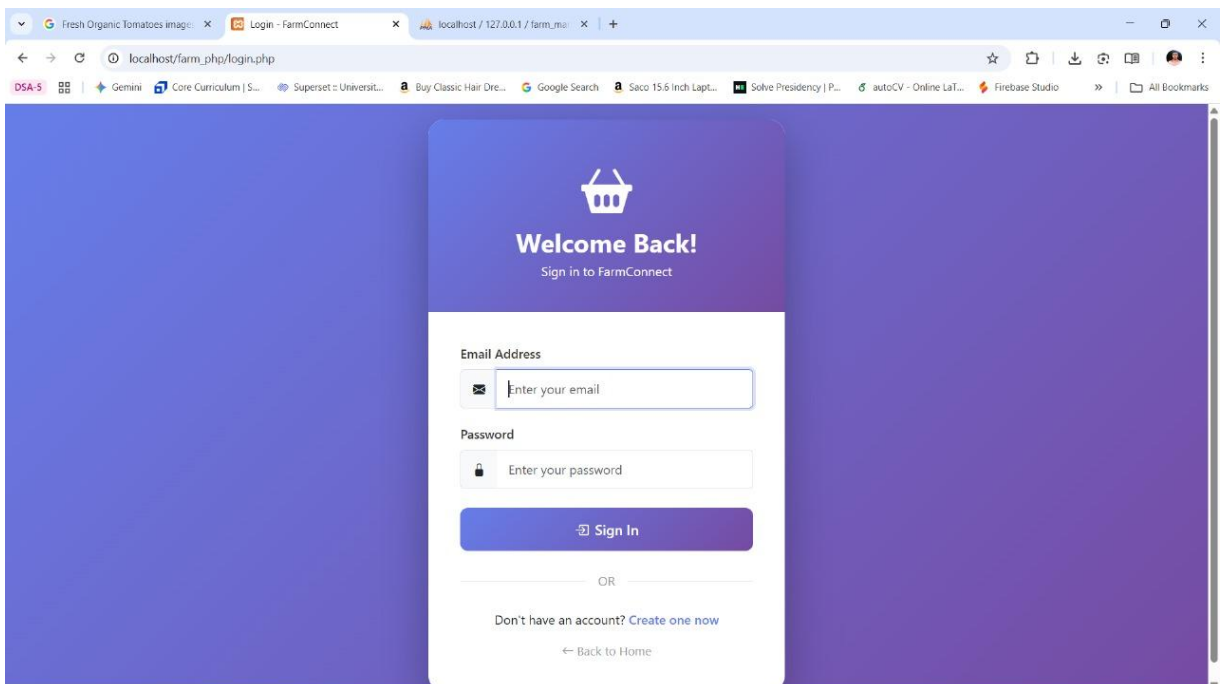


Fig A.3 Login Page

3. Shop Page

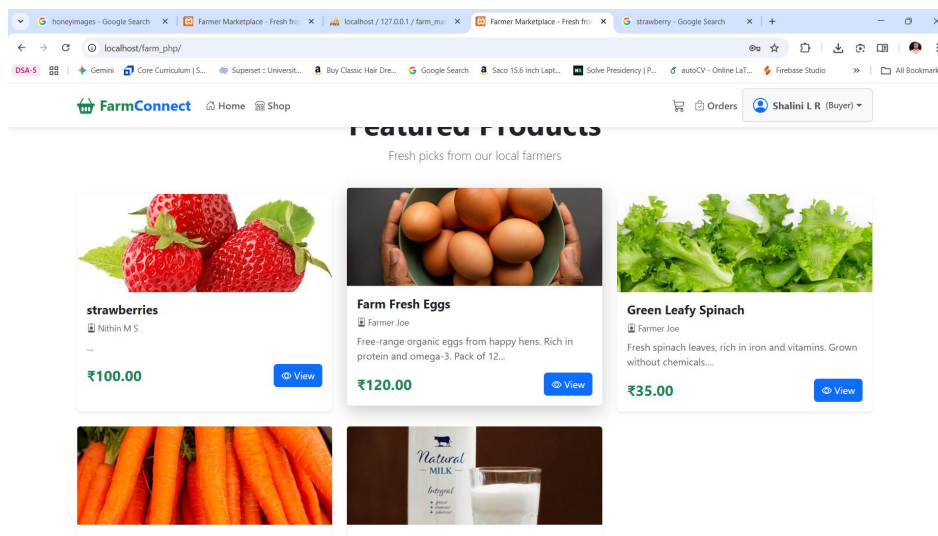


Fig A.4 Shop Page

4. Cart Page

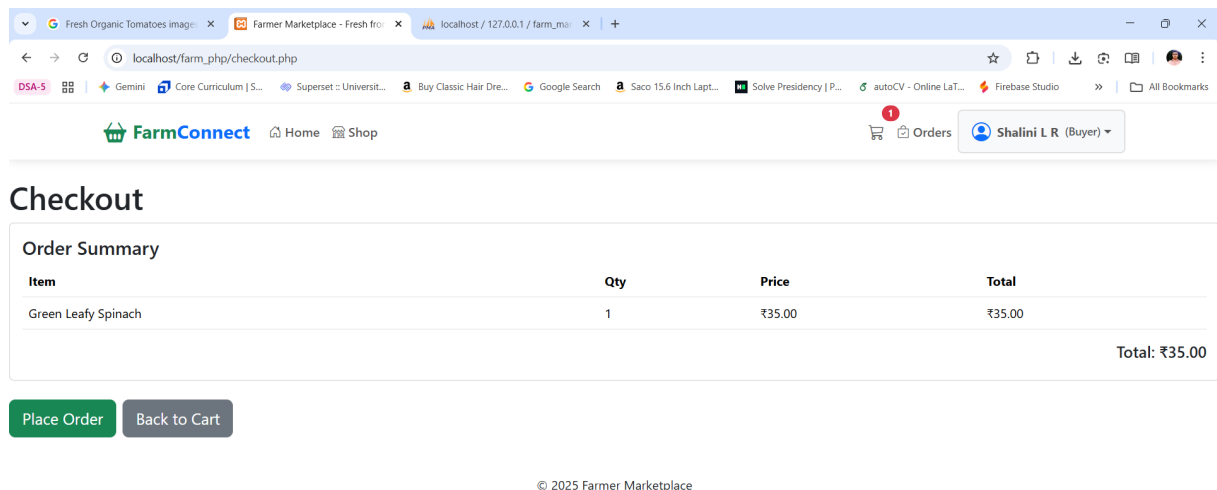


Fig A.5 Cart Page

5. Farmer Add Product Page

The screenshot shows the 'Add New Product' page of the FarmConnect web application. The page has a header with the FarmConnect logo, navigation links (Home, Shop), and user information (Nithin M S (Farmer)). The main content area is titled 'Add New Product' and contains a form with the following fields:

- Product Name ***: A text input field with the placeholder 'e.g., Fresh Tomatoes'.
- Category**: A dropdown menu with the placeholder 'Select Category'.
- Description**: A text area with the placeholder 'Describe your product...'.
- Price (₹) ***: A text input field with the value '0.00'.
- Quantity ***: A text input field with the value '0'.
- Product Image**: A file upload section with a 'Choose File' button, 'No file chosen' text, and a note 'Max size: 10MB. Formats: JPG, PNG, GIF, WebP'.

At the bottom of the form are two buttons: 'Add Product' (green) and 'Cancel' (grey).

Fig A.6 Farmer Add Product Page

6. Farmer Dashboard Page

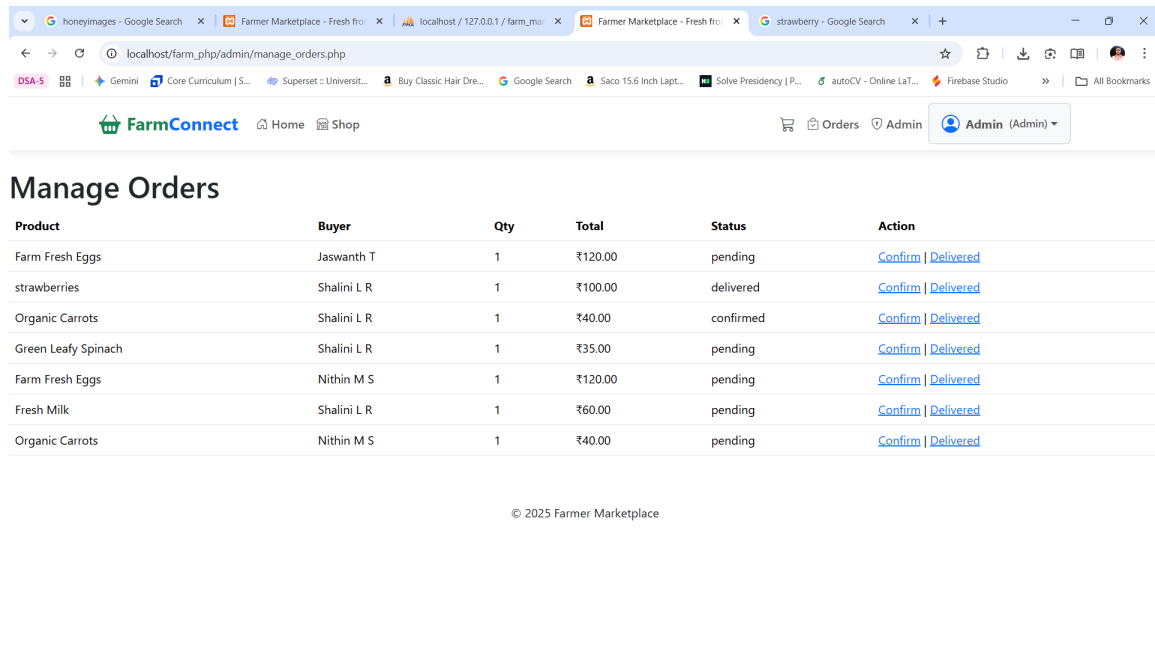
The screenshot shows the 'Farmer Dashboard' page of the FarmConnect web application. The page has a header with the FarmConnect logo, navigation links (Home, Shop), and user information (Nithin M S (Farmer)). The main content area is titled 'Farmer Dashboard' and includes a welcome message 'Welcome back, Nithin M S!'. There is a green 'Add New Product' button in the top right corner. Below the header, there are four summary cards:

- TOTAL PRODUCTS**: 0 (with a box icon)
- TOTAL STOCK**: 0 (with a box icon)
- TOTAL ORDERS**: 0 (0 pending) (with a shopping cart icon)
- TOTAL REVENUE**: ₹0.00 (with a rupee symbol icon)

Below the summary cards is a 'Quick Actions' section with three buttons: '+ Add Product' (green), 'View Orders' (blue), and 'View Shop' (grey). At the bottom, there is a 'My Products' section with a placeholder icon of a box.

Fig A.7 Farmer Dashboard Page

7. Admin Manage Order Page



Product	Buyer	Qty	Total	Status	Action
Farm Fresh Eggs	Jaswanth T	1	₹120.00	pending	Confirm Delivered
strawberries	Shalini L R	1	₹100.00	delivered	Confirm Delivered
Organic Carrots	Shalini L R	1	₹40.00	confirmed	Confirm Delivered
Green Leafy Spinach	Shalini L R	1	₹35.00	pending	Confirm Delivered
Farm Fresh Eggs	Nithin M S	1	₹120.00	pending	Confirm Delivered
Fresh Milk	Shalini L R	1	₹60.00	pending	Confirm Delivered
Organic Carrots	Nithin M S	1	₹40.00	pending	Confirm Delivered

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Fig A.8 Admin Manage Order Page

4. Auxiliary Documents

- API credentials are placed in a configuration file (e.g., config.js) demonstrating how API keys are securely stored and can be used.
- Project plan / design document outlining the system architecture.
- Test cases used to verify functionality for each category.

Similarity Check for Report

Babitha S-Report_final_PSCS281.pdf

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